

An asymptotic formula for the number of totients

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September 25, 2026

Abstract

Let $V(x)$ count the distinct values of Euler's totient function up to x . We give an explicit asymptotic equivalent for $V(x)$. Its coefficient is a uniform limit of functions defined from finite arithmetic data. We also prove that $V(cx)/V(x) \rightarrow c$ as $x \rightarrow \infty$ for every fixed $c > 0$, answering a question of Erdős and Hall.

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1 Introduction

Euler's totient function $\varphi(n)$ counts the integers in $\{1, \dots, n\}$ that are relatively prime to n . We write

$$\mathcal{V} = \{\varphi(n) : n \geq 1\}, \quad V(x) = \#\{v \in \mathcal{V} : v \leq x\}.$$

Counting distinct values requires more than counting integers with a prescribed factorization: different integers can have the same totient. This distinction is decisive in an asymptotic formula for $V(x)$.

Throughout the paper logarithms are natural, and $\log_j$ denotes the j -fold iterated logarithm.

The counting problem. Since $\varphi(p) = p - 1$ for primes p , the prime number theorem gives $V(x) \geq \pi(x+1) \sim x/\log x$. Pillai proved that the totient values have density zero [12]. Erdős used the normal number of prime factors of $p - 1$ to sharpen the upper bound to $V(x) \ll_\varepsilon x/(\log x)^{1-\varepsilon}$ for every $\varepsilon > 0$ [1, Section 2]. His later count of distinct values $\varphi(pr)$ with p, r prime and $pr \leq x$ gives the lower bound $(1 + o(1))x \log_2 x / \log x$ [2, pp. 542–543]. Erdős and Hall obtained a much larger factor $\exp\{a(\log_3 x)^2\}$ for every $0 < a < 1/\log 16$ [6, pp. 1–3]. Pomerance brought the upper bound onto the same logarithmic scale [14, Equation (1.4)]. Maier and Pomerance then determined the leading constant [10, Section 1]:

$$V(x) = \frac{x}{\log x} \exp\{(C_0 + o(1))(\log_3 x)^2\}, \quad C_0 = 0.8178146 \dots$$

In 1998, Ford refined this to an estimate with only a bounded multiplicative uncertainty [7, Theorem 1]:

$$V(x) = \frac{x}{\log x} \exp \left\{ \begin{array}{l} C_0(\log_3 x - \log_4 x)^2 + D_0 \log_3 x \\ - (D_0 + \frac{1}{2} - 2C_0) \log_4 x + O(1) \end{array} \right\}, \quad (1.1)$$

where $D_0 = 2.1769687 \dots$; both constants are specified in the notation of Section 2 below. Ford also described the typical prime factorizations of preimages [7, Theorems 10 and 16], which supply the structural starting point of our proof. All numbered references to his distribution paper use the revised arXiv version of 14 July 2013.

Fixed dilation. Erdős and Hall asked whether

$$\frac{V(cx)}{V(x)} \longrightarrow c \quad (c > 1 \text{ fixed})$$

in the final paragraph of [6, p. 3]. Erdős repeated the question in [4, p. 80]. Ford proved that $V(cx) - V(x) \asymp_c V(x)$ for every fixed $c > 1$ [7, Theorem 4]. This controls the number of values in a fixed multiplicative interval, but neither it nor (1.1) determines the limiting ratio.

We obtain an explicit asymptotic equivalent and prove the fixed-scale limit, thereby answering this question of Erdős and Hall positively. The coefficient in the equivalent retains finite arithmetic information. It is defined in Section 2 by bounded prime and integer sums, inclusion–exclusion, and a uniformly convergent limit; its definition does not use V .

The source of the coefficient. Order the prime factors of a typical preimage from largest to smallest, and separate a long prefix from a shorter tail. Ford's normal-structure estimates put the double logarithms of the large primes in a simplex. Most of that simplex can be counted by volume. Near its last coordinates the primes remain discrete, however, and several tail factorizations may

give the same totient. We retain these tails exactly and take the volume of the union of the regions that they permit. Inclusion–exclusion of this finite union produces the arithmetic coefficient.

Erdős’s semiprime argument already controls collisions between products of shifted primes [2, Lemma 1]. The quantitative shifted-prime normality and layered comparison used here were developed by Maier and Pomerance [10, Sections 2–3] and Ford [7, Section 5]. Here the comparison is proved under the hypotheses needed for the long prefix, with explicit costs for recovering the tuple data inside the residual factors. The resulting control of all ordered prefix collisions, combined with the exact finite-tail union, converts the volume into a count of distinct values. Counting the largest prime with common endpoint parameters first gives regular variation. Identifying the resulting mass with the arithmetic coefficient then gives the explicit equivalent. The errors are uniform in the phase; neither conclusion assumes continuity of the coefficient.

The prefix uniqueness used here is not uniqueness of the entire preimage: different arithmetic tails may still give the same value. A separate result on extreme complete fibers gives, for every fixed $\varepsilon > 0$, infinitely many positive integers v with $\#\{n \geq 1 : \varphi(n) = v\} > v^{1-\varepsilon}$ [11, Theorem 1.1]. That result concerns large individual fibers, rather than the asymptotic number of distinct values, and is not an input to the present proof.

Least preimages. For $v \in \mathcal{V}$, let

$$\ell(v) = \min\{n \geq 1 : \varphi(n) = v\}.$$

For each fixed positive integer k , the same passage of Erdős [4, p. 80] asks about the number of values $v \leq x$ for which $kx < \ell(v) \leq (k+1)x$. Section 7 gives a weighted asymptotic and proves that this count has order $V(x)$ precisely when some totient d satisfies $\ell(d) > kd$; otherwise the count vanishes identically. The positive alternative holds for $k = 1, 2$. We do not determine which other integers k satisfy this seed condition. In particular, the separate question whether $\ell(d)/d$ is unbounded, posed in [5, Section I.9], remains unresolved here.

2 The explicit main term

We first give the counting scale and the main theorem, then construct its arithmetic coefficient. Definitions involving x are used only for sufficiently large x .

2.1 The scale

For $j \geq 1$, put

$$a_j = (j+1) \log(j+1) - j \log j - 1 = \int_j^{j+1} \log t \, dt. \quad (2.1)$$

These numbers are positive. The increasing function $\sum_{j \geq 1} a_j z^j$ goes from zero to infinity on $0 < z < 1$, so there is a unique $\rho \in (0, 1)$ such that

$$\sum_{j \geq 1} a_j \rho^j = 1.$$

These coefficients and this root occur in Maier and Pomerance’s asymptotic estimate [10, Section 1]. Define

$$\lambda = \log(1/\rho), \quad \gamma = \left(\sum_{j \geq 1} j a_j \rho^j \right)^{-1}, \quad g_0 = 1, \quad g_j = \sum_{d=1}^j a_d g_{j-d}. \quad (2.2)$$

Our γ is the constant denoted by λ in Ford's recurrence estimates [7, Section 3]. Set

$$C_0 = \frac{1}{2\lambda}, \quad D_0 = \frac{1 + \log(\lambda/\gamma)}{\lambda} - \frac{1}{2};$$

these are the constants in the classical estimate (1.1). For the scale used in our theorem, put

$$B = \log_2 x, \quad m = \left\lfloor \frac{\log B - \log_2 B}{\lambda} \right\rfloor, \quad \theta = \frac{\log B - \log_2 B}{\lambda} - m, \quad (2.3)$$

so that $0 \leq \theta < 1$, and write

$$G_j = \frac{B^j}{j! \prod_{i=1}^j g_i} \quad (0 \leq j \leq m), \quad G_0 = 1. \quad (2.4)$$

The factor $xG_m/\log x$ is Ford's counting scale. The following theorem resolves its bounded multiplicative uncertainty by an explicitly constructed function of the phase θ .

Theorem 2.1. *For sufficiently large positive integers H , there are nonnegative functions $A_H(1; \cdot) : [0, 1) \rightarrow \mathbb{R}$, given by the finite arithmetic formula (2.8) below, which converge uniformly as $H \rightarrow \infty$ to a function $A(1; \cdot)$ satisfying*

$$0 < \inf_{0 \leq s < 1} A(1; s) \leq \sup_{0 \leq s < 1} A(1; s) < \infty.$$

With the parameters (2.2)–(2.4),

$$V(x) \sim \frac{x}{\log x} G_m A(1; \theta).$$

Moreover, for every fixed real $c > 0$,

$$\frac{V(cx)}{V(x)} \rightarrow c.$$

The first argument of A records a weight; the main theorem uses the constant weight 1. Its arithmetic construction below does not use V . The fixed-scale conclusion has a separate mechanism: it compares the same representation count at two endpoints. No regularity of $s \mapsto A(1; s)$ is asserted or needed.

2.2 The arithmetic coefficient

The tail index runs in the opposite direction from the ordered prime index. In the proof we write a preimage as $p_0 p_1 \cdots p_L a$, with the primes decreasing, and use two cuts

$$R = m - H, \quad L = m - P, \quad P = \lfloor \log \log H \rfloor.$$

The long prefix ends at p_R , whereas the exact arithmetic tail contains $p_{R+1}, \dots, p_L$ and a . Thus the notation below is $Q_h = p_{m-h}$: its indices run from P to $H - 1$, and Q_P is the smallest retained prime. For fixed H , these tail primes are bounded independently of x . We first let x tend to infinity with H fixed, and only afterwards let H tend to infinity. The following definition uses the phase s directly, without reference to x .

Fix $s \in [0, 1)$ and put $\alpha_s = \lambda e^{\lambda s}$. Let H be a sufficiently large positive integer and $P = \lfloor \log \log H \rfloor$. A *tail witness* is a choice

$$\eta = ((Q_h)_{P \leq h < H}, a)$$

of primes Q_h and a positive integer a with the following properties. Writing $v_h = \log_2 Q_h$, we require

$$\begin{aligned} 0.9\alpha_s h \rho^{-h} &\leq v_h \leq 1.1\alpha_s h \rho^{-h} & (P \leq h < H), \\ \sum_{l=P}^{h-1} a_{h-l} v_l &\leq (1 + 10^{-4} e^{-h/40}) v_h & (P \leq h < H), \\ P^+(a) &\leq Q_P, \quad \log a \leq \exp(2\alpha_s P \rho^{-P}). \end{aligned} \tag{2.5}$$

Here $P^+(a)$ is the largest prime factor of a , with $P^+(1) = 1$; an empty sum is zero. Associate to η the integer

$$w(\eta) = a \prod_{h=P}^{H-1} Q_h.$$

For $d \in \mathcal{V}$, let $\mathcal{W}_{H,s}(d)$ be the set of tail witnesses for which $\varphi(w(\eta)) = d$. For $h \geq H$, define

$$D(h, \eta) = \sum_{l=P}^{H-1} a_{h-l} v_l. \tag{2.6}$$

The set of all possible witnesses, even as s varies, is finite for each fixed H : α_s lies between λ and λ/ρ , and every prime and integer in (2.5) is consequently bounded in terms of H . Also $D(h, \eta) = O_H(\log(h+1))$. Thus the series used below converges absolutely, uniformly over this finite set and over s .

A tail totient must contribute once even when it has several witnesses. The union that enforces this can be described by independent exponential random variables E_h of mean one, indexed by $h \geq H$. For each witness put

$$\mathcal{E}_\eta = \bigcap_{h \geq H} \left\{ E_h \geq \frac{\gamma}{\alpha_s} \rho^h D(h, \eta) \right\}. \tag{2.7}$$

For a function $f : [1, \infty) \rightarrow [0, 1]$, the coefficient is

$$A_H(f; s) = \rho^{H(H-1)/2} \left(\frac{\gamma}{\alpha_s} \right)^H \sum_{\substack{d \in \mathcal{V} \\ \mathcal{W}_{H,s}(d) \neq \emptyset}} \frac{f(\ell(d)/d)}{d} \Pr \left(\bigcup_{\eta \in \mathcal{W}_{H,s}(d)} \mathcal{E}_\eta \right).$$

An intersection of events requires each E_h to exceed the largest of the corresponding thresholds. Independence and absolute convergence of their sum therefore turn finite inclusion–exclusion into the explicit expression

$$\begin{aligned} A_H(f; s) &= \rho^{H(H-1)/2} \left(\frac{\gamma}{\alpha_s} \right)^H \sum_{\substack{d \in \mathcal{V} \\ \mathcal{W}_{H,s}(d) \neq \emptyset}} \frac{f(\ell(d)/d)}{d} \\ &\quad \times \sum_{\emptyset \neq T \subseteq \mathcal{W}_{H,s}(d)} (-1)^{|T|-1} \exp \left\{ -\frac{\gamma}{\alpha_s} \sum_{h=H}^{\infty} \rho^h \max_{\eta \in T} D(h, \eta) \right\}. \end{aligned} \tag{2.8}$$

For $f = 1$, the factor involving ℓ is simply 1: no least-preimage search is part of the main coefficient. For other weights, any witness with totient d supplies the finite search bound $\ell(d) \leq w(\eta)$. The probability expression also shows directly that $A_H(f; s) \geq 0$. Section 6 identifies these probabilities

with limiting normalized prefix volumes: $1/d$ comes from counting the largest prime, and the prefactor is the limiting ratio of the prefix volume to G_m .

When the limit exists, write

$$A(f; s) = \lim_{H \rightarrow \infty} A_H(f; s).$$

Each approximant uses finite arithmetic data and an absolutely convergent series, not an unspecified counting constant. Theorem 2.1 justifies the limit for $f = 1$; the next result does so for the weights needed to count least preimages. Convergence for arbitrary f is not required.

2.3 Least-preimage intervals

For a positive integer k , define

$$\begin{aligned} N_k(x) &= \#\{v \in \mathcal{V} : v \leq x, \ kx < \ell(v) \leq (k+1)x\}, \\ f_k(r) &= \min\{1, (k+1)/r\} - \min\{1, k/r\}. \end{aligned} \tag{2.9}$$

The function f_k is nonnegative and bounded by 1.

Theorem 2.2. *For each fixed positive integer k , the functions $A_H(f_k; s)$ converge uniformly for $0 \leq s < 1$, and*

$$N_k(x) = \frac{x}{\log x} G_m(A(f_k; \theta) + o(1)).$$

There are two possibilities.

(i) *If some $d \in \mathcal{V}$ satisfies $\ell(d) > kd$, then*

$$\inf_{0 \leq s < 1} A(f_k; s) > 0, \quad N_k(x) \sim \frac{x}{\log x} G_m A(f_k; \theta) \asymp_k V(x).$$

(ii) *If no such d exists, then $N_k(x) = 0$ for every $x > 0$, and $A(f_k; s) = 0$ for every $s \in [0, 1)$.*

The first alternative holds for $k = 1, 2$.

This theorem does not determine which other integers k satisfy the first alternative. Whether the first alternative holds for every positive integer k is equivalent to the unboundedness of $\ell(d)/d$; the weighted formula does not settle this question.

2.4 Proof strategy

Two parts of the argument ensure that this construction counts values once. First, outside an exceptional set, *every* preimage has the required form. Second, collisions between different long prefixes have negligible total multiplicity. For the latter we use the layered shifted-prime method of Maier and Pomerance [10, Section 3], in the form developed by Ford, and adapt his comparison [7, Lemma 5.1] to a different set of hypotheses, proving the necessary uniform bound in Section 4. The number of layers grows, but each sieve application concerns only two or three linear forms. The comparison and the subsequent control of tuple multiplicities are the ingredients that turn a volume estimate into an asymptotic for distinct values.

The end of the continuous prefix and the end of the retained prime list are different cuts. Their separation makes the exceptional-set estimates small enough to absorb the smooth tail. Section 3 obtains the required preimages directly from Ford's Theorems 10 and 16 and develops

the volume estimates. Section 5 proves prefix uniqueness and transfers tuple counts to distinct values. Section 6 counts the largest prime using common endpoint parameters, obtaining fixed dilation before identifying the arithmetic limit. Finally, Section 7 weights that prime count to locate least preimages and proves the positive and zero alternatives.

3 Extracting a long prime prefix

Our first task is to represent almost every totient by a long prime prefix and a bounded arithmetic tail. The exceptional set must be small in a stronger sense: outside it, *every* preimage must have the required form. Later we shall count distinct prefix–tail tuples, and remove tuples that fail the conditions needed to compare two representations.

Throughout this section H is a sufficiently large fixed integer, $P = \lfloor \log \log H \rfloor$, and

$$R = m - H, \quad L = m - P. \quad (3.1)$$

We first let $x \rightarrow \infty$, and then let $H \rightarrow \infty$. Write $h_i = m - i$ and

$$\alpha = \frac{B\rho^m}{m}, \quad b_i = \alpha h_i \rho^{-h_i}, \quad \tilde{b}_i = \alpha_\theta h_i \rho^{-h_i}, \quad \xi_i = 1 + 10^{-4} e^{-h_i/40}.$$

In particular $b_0 = B$. The definitions in (2.3) give

$$\frac{\alpha}{\alpha_\theta} = \frac{\log B}{\lambda m} = 1 + \frac{\lambda\theta + \log \log B}{\lambda m} = 1 + O\left(\frac{\log m}{m}\right). \quad (3.2)$$

Thus α and α_θ stay in a fixed compact subinterval of $(0, \infty)$, uniformly in the phase. We shall only need the coarse bounds

$$0.54 < \rho < 0.545, \quad (3.3)$$

which follow from the numerical value recorded in [7, (1.4)]. They imply $\lambda < 1$, $18 < 30\lambda < 20$, and $1.1\rho < 0.6$. Together with (3.2), these inequalities justify the fixed constants 0.61 and 0.62 used below. We use

$$K_H = \exp(CP\rho^{-P}) = H^{o(1)},$$

where the absolute constant C may be enlarged from one upper bound to the next. The last equality follows from $\lambda = \log(1/\rho) < 1$. The notation ε_H denotes a positive quantity tending to zero with H , and $o_{x;H}(1)$ denotes a quantity tending to zero as $x \rightarrow \infty$ for fixed H . All such estimates below are uniform in the phase.

3.1 Structure estimates from Ford

Ford's order estimate and fundamental-simplex estimates give

$$V(x) \asymp \frac{x}{\log x} G_m, \quad (3.4)$$

and his recurrence estimate gives

$$g_j = \gamma \rho^{-j} + O(1), \quad g_j \asymp \rho^{-j}. \quad (3.5)$$

See [7, Theorem 1, Lemmas 3.4 and 3.7, and Corollary 3.5]. Ford denotes our constant γ by λ . In particular, for $0 \leq j \leq m$,

$$\frac{G_j}{G_m} = \prod_{i=j+1}^m \frac{ig_i}{B} \leq C^{m-j} \rho^{(m-j)(m-j-1)/2} \ll 1. \quad (3.6)$$

We shall use two quantitative structure results of Ford. Order the prime factors of a preimage n with repetitions as $p_0 \geq p_1 \geq \dots$. For each fixed $0 < \eta \leq 1/8$, his Theorem 10, together with the missing-prime exception in Theorem 16, implies

$$\# \left\{ v \leq x : \begin{array}{l} v \in \mathcal{V}, \text{ some preimage has} \\ |\log_2 p_i/b_i - 1| > \eta \text{ for some } 1 \leq i \leq m - P \end{array} \right\} \ll_{\eta} V(x) e^{-c_{\eta} P}. \quad (3.7)$$

Here and below a missing indicated prime is a failure of the condition. For completeness, the summation giving (3.7) is short. For $i \leq 3\eta m$, Theorem 10(b) contributes at most $Cme^{-\eta m/13} V(x)$. For $i > 3\eta m$, Theorem 10(a) contributes at most

$$CV(x) \frac{i}{\eta m} \exp \left\{ -\frac{\eta^2 m(m-i)}{4i} \right\} \leq \frac{CV(x)}{\eta} e^{-\eta^2(m-i)/4}.$$

Summing over $m - i \geq P$ proves the assertion for sufficiently large x when the indicated primes exist. A missing p_L is included in the $O(V(x)e^{-P^2/4})$ exceptional set of Theorem 16, which is absorbed in the displayed bound. The convention preceding Ford's Theorem 10 counts values with *some* failing preimage; hence the complement of this set controls every preimage.

Ford's Theorem 16, with its parameter $\Psi = P$, states that all but $O(V(x)e^{-P^2/4})$ values have every preimage satisfying

$$\log_2 p_L > 2, \quad \sum_{r=i+1}^L a_{r-i} u_r \leq \xi_i u_i \quad (0 \leq i < L), \quad u_0 = B, \quad u_i = \log_2 p_i \quad (i \geq 1). \quad (3.8)$$

Its final adjacent-coordinate inequality has coefficient 1 in place of $a_1 < 1$; weakening that inequality gives (3.8).

Write $\Omega(n, U, T)$ for the number of prime factors of n belonging to $(U, T]$, counted with multiplicity, and put $\Omega(n) = \Omega(n, 1, n)$, including $\Omega(1) = 0$.

Definition 3.1. For $S \geq e^e$, a prime p is S -normal if

$$\Omega(p-1, 1, S) \leq 2 \log_2 S$$

and, for every $S \leq U < T \leq p-1$,

$$|\Omega(p-1, U, T) - (\log_2 T - \log_2 U)| < \sqrt{\log_2 S \log_2 T}.$$

We use Definition 1 in the 2013 revision [7]. Its Lemma 2.6 gives, uniformly for $z \geq 3$,

$$\#\{p \leq z : p \text{ is not } S\text{-normal}\} \ll \frac{z}{\log z} (\log_2 z)^5 (\log S)^{-1/6}. \quad (3.9)$$

The function Ω in these statements counts prime factors with multiplicity.

3.2 Basic witnesses and the coverage target

We now specify the representations that these structure estimates will provide. The simplex notation records the inequalities on their prime coordinates; the bounds on the last factor will make the arithmetic tail finite for fixed H .

For integers $1 \leq j \leq m$, $u_0 = B$, and $\beta_i \geq 1$ for $0 \leq i < j$, put $\beta_j = 1$ and define the simplex

$$S_j(\beta) = \left\{ (u_1, \dots, u_j) : s_i := \beta_i u_i - \sum_{r=i+1}^j a_{r-i} u_r \geq 0 \quad (0 \leq i \leq j) \right\}.$$

The terminal inequality is $u_j \geq 0$. For $j = 0$, the simplex consists of the empty tuple and has volume $G_0 = 1$.

Definition 3.2. A *basic witness* is a choice of primes $p_0, \dots, p_L$ and an integer $a \geq 1$ satisfying

$$\begin{aligned} p_0 &\geq x^{0.9}, & 0.9\tilde{b}_i &\leq \log_2 p_i \leq 1.1\tilde{b}_i \quad (1 \leq i \leq L), \\ (u_1, \dots, u_L) &\in S_L(\xi), & P^+(a) &\leq p_L, & \log a &\leq \exp(2\tilde{b}_L). \end{aligned} \quad (3.10)$$

Here $u_0 = B$, and $u_i = \log_2 p_i$ for $i \geq 1$.

For our parameter range the prime bands are disjoint and decreasing; also $p_0 > p_1$. Thus the displayed prime prefix is strictly decreasing. Repetitions of p_L inside a are allowed.

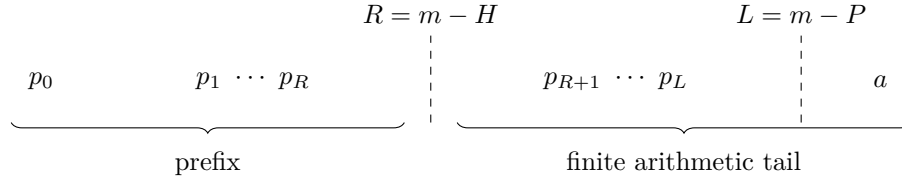


Figure 1: The two cuts in a basic witness. We count p_0 by the prime number theorem and approximate the $p_1, \dots, p_R$ coordinates by volume. The later primes and the smooth integer a remain discrete.

Proposition 3.3. *There is a set $E_x \subseteq \mathcal{V} \cap [1, x]$ with*

$$|E_x| \leq \{\varepsilon_H + o_{x;H}(1)\} V(x)$$

such that every preimage of every $v \in (\mathcal{V} \cap [1, x]) \setminus E_x$ is $p_0 \cdots p_L a$ for a basic witness. One may take

$$\varepsilon_H \ll e^{-cP} + K_H \exp\{-\exp(cP\rho^{-P})\}.$$

For every basic witness, putting $w = p_{R+1} \cdots p_L a$, we have

$$\begin{aligned} p_1 \cdots p_L a &\leq \exp((\log x)^{0.8}), \\ \log w &\leq H \exp(0.62b_R) + \exp(2\tilde{b}_L), & w &< p_R. \end{aligned} \quad (3.11)$$

In particular w is bounded in terms of H alone, uniformly in the phase.

We prove the proposition after the volume and smooth-cofactor estimates below. Those estimates also prepare a second task: we shall retain the shortened data $(p_0, \dots, p_R, \varphi(w))$, then delete those that fail the strict inequalities and normality conditions needed for the collision argument. Coverage concerns *values*; this later deletion must count the distinct shortened tuples, including all tuples that represent the same value.

3.3 Simplex volume and concentration

The coordinate scaling follows Ford's fundamental-simplex method [7, Lemma 3.2 and Corollary 3.3]. Our last adjacent coefficient is a_1 rather than 1, so the exact volume is slightly different from his; we derive it from the slacks below. The recurrence estimate (3.5) is Ford's Lemma 3.7; his Remark 1 credits its streamlined proof to Lau's solution of a recurrence problem [9].

The next estimates turn sums over the allowed prime coordinates into simplex volumes. We also bound the mass near a coordinate boundary or a simplex face, so that removing these regions has a controlled cost.

Lemma 3.4. *Let $1 \leq j \leq m$. The true simplex $S_j(\mathbf{1})$ has volume G_j . Under its uniform volume measure the variables $W_i = g_i s_i / B$ are uniform on $\{W_i \geq 0 : \sum_{i=0}^j W_i = 1\}$. Moreover,*

$$\text{Vol } S_j(\beta) \leq G_j \prod_{r < j} \beta_r^{j-r}.$$

Within $S_j(\beta)$, a specified slack s_t in an interval of length D occupies at most a fraction $j g_t D / B$ of the volume. A specified coordinate u_t , $1 \leq t \leq j$, in such an interval occupies at most $j g_t \beta_t D / B$.

Proof. For the true simplex the triangular inversion is

$$u_i = \sum_{t=i}^j g_{t-i} s_t, \quad B = \sum_{t=0}^j g_t s_t.$$

The transformation from $u_1, \dots, u_j$ to $s_1, \dots, s_j$ has determinant 1. This proves the volume and distribution statements.

Set $K_i = \prod_{r < i} \beta_r$, $K_0 = 1$. The map $u_i \mapsto u_i / K_i$ sends $S_j(\beta)$ into $S_j(\mathbf{1})$, because $K_r \geq \beta_i K_i$ when $r > i$. Its inverse Jacobian is $\prod_{r < j} \beta_r^{j-r}$.

For the slice estimates, define $\gamma'_0 = 1$ and $\gamma'_t = \beta_t^{-1} \sum_{i < t} \gamma'_i a_{t-i}$. Then $0 < \gamma'_t \leq g_t$ and

$$\sum_{t=0}^j \gamma'_t s_t = \beta_0 B.$$

The marginal density of s_t , under the uniform measure on this simplex, is at most $j \gamma'_t / (\beta_0 B)$. For the coordinate assertion, fix all other positive-index slacks. The coordinate u_t is affine in s_t with slope $1/\beta_t$. Integrating first in this fiber and bounding by the full projection volume gives the additional factor β_t . $\square$

The following concentration estimate will be used in Section 6 to remove the prefix bands after replacing prime sums by volumes. Write $z_+ = \max\{z, 0\}$.

Lemma 3.5. *Fix $\eta > 0$. There are constants $C_\eta, c_\eta > 0$ such that, if Q is an integer with $0 \leq Q < m$, $j = m - Q$, $1 \leq i \leq j$, and $h_i \geq C_\eta(Q + 1)$, then uniform volume measure on $S_j(\mathbf{1})$ satisfies*

$$\Pr(|u_i/b_i - 1| > \eta) \leq C_\eta e^{-c_\eta h_i}.$$

Also, for $T \geq 0$,

$$\Pr(u_j \geq T) = (1 - g_j T / B)_+^j \leq e^{-j g_j T / B}.$$

Proof. By (3.5) and Lemma 3.4,

$$\frac{u_i}{B\rho^i} = \sum_{t=i}^j (1 + O(\rho^{t-i})) W_t.$$

Represent $W_t = E_t / \sum_{r=0}^j E_r$, where the E_t are independent mean-one exponential variables. The main numerator has $j - i + 1 = h_i - Q + 1$ terms, and

$$\frac{(j - i + 1)/(j + 1)}{h_i/m} = 1 + O((Q + 1)/h_i).$$

Exponential moment bounds give exponentially small probabilities for a fixed relative deviation of either the numerator or the denominator. The error numerator is bounded in absolute value by $C \sum_{r \geq 0} \rho^r E_{i+r}$; this sum has a uniformly bounded exponential moment, since $\prod_{r \geq 0} (1 - z\rho^r)^{-1} < \infty$ for $0 < z < 1$. Once $h_i \geq C_\eta(Q + 1)$, these three bounds give the first assertion. The second is the exact marginal tail of W_j . $\square$

3.4 Prime boxes and smooth cofactors

The use of reciprocal prime sums and enlarged coordinate boxes follows Ford's Lemma 3.1 and the thickening arguments of his Section 3 [7]. We record the quantitative bounds for our simplexes, including the boundary errors needed for the tail unions.

We use unit boxes in the coordinates $\log_2 p_i$. Mertens' theorem gives

$$\sum_{b \leq \log_2 p < b+1} \frac{1}{p-1} = 1 + O(e^{-b}) \quad (b \geq 2). \quad (3.12)$$

The replacement of $1/p$ by $1/(p-1)$ changes the sum by $O(\exp(-\exp b))$. When the lower endpoints grow geometrically backwards through the coordinates, multiplying (3.12) costs only a bounded factor; if their sum of errors is small, it is the relative product error.

Lemma 3.6. *Let $0 \leq j \leq L$. Sum over primes $p_1, \dots, p_j$ whose coordinates satisfy*

$$0.9\tilde{b}_i \leq u_i \leq 1.1\tilde{b}_i, \quad (u_1, \dots, u_j) \in S_j(\boldsymbol{\xi}).$$

Then

$$\sum_{p_1, \dots, p_j} \prod_{i=1}^j \frac{1}{p_i - 1} \ll G_m. \quad (3.13)$$

Every unit box meeting this region lies in $S_j(\boldsymbol{\beta})$, where, with a sufficiently large absolute constant,

$$\beta_t = 1 + C(e^{-h_t/40} + h_t^2 \rho^{h_t}) \quad (0 \leq t < j), \quad \beta_j = 1. \quad (3.14)$$

This enlargement has bounded volume cost. Its coordinate scaling factors through R differ from 1 by $O(e^{-H/40} + H^2 \rho^H)$.

There is also the following shell estimate in L coordinates. Fix $A > 0$. Among unit boxes meeting the displayed region, retain those with a point satisfying, for some $i \leq R$, at least one of these conditions:

- (i) $i \geq 1$ and u_i is within A of a coarse band endpoint;

(ii) the absolute value of $\xi_i u_i - \sum_{r=i+1}^L a_{r-i} u_r$ is at most $A h_i^2$;

(iii) $\sum_{r=i+1}^L a_{r-i} u_r \geq (1 - h_i^{-4}) u_i$.

The total volume of these boxes, and their total reciprocal-prime weight, is

$$O_A(H^{-2} G_m). \quad (3.15)$$

The same estimate holds when any subset of coordinates uses reciprocal-prime measure and the others use Lebesgue measure. For conditions (i)–(ii) alone, or for the region lost by replacing the prefix coefficients ξ_i , $0 \leq i \leq R$, with 1, the stronger bound is

$$O_A \left(G_m \sum_{h \geq H} (h e^{-h/40} + h^3 \rho^h) \right).$$

Proof. The band lower bounds are $u_t \gg h_t \rho^{-h_t}$. Changing each coordinate by at most 1 changes inequality t by $O(h_t^2)$. These errors are absorbed by (3.14), including at $t = 0$, where $u_0 = B$ is fixed. The logarithm of its Jacobian upper bound is at most

$$C \sum_{h \geq P} h (e^{-h/40} + h^2 \rho^h) \ll 1.$$

Lemma 3.4, (3.6), and (3.12) prove (3.13). The coordinate scaling factor is $\prod_{t < i} \beta_t$, without the Jacobian exponents; summing for $h_t > h_i \geq H$ proves its stated bound.

For the shells put $h = h_i$. Throughout a box meeting the bands, $u_i \ll b_i$. Conditions (ii) and (iii), together with the unit perturbations, put the enlarged slack in an interval starting at zero of length at most

$$C_A \{b_i (h^{-4} + e^{-h/40}) + h^3\}.$$

Here the h^3 term includes $h^2 \rho^h u_i = O(h^3)$. Since $L g_i / B \ll \rho^h$, Lemma 3.4 bounds the relative volume by

$$C_A (h^{-3} + h e^{-h/40} + h^3 \rho^h).$$

Condition (i), after expanding its interval by 2, costs only $O_A(\rho^h)$. Sum these estimates over $h \geq H$, and use the bounded enlargement cost. The prime-weight assertion follows again from (3.12). For the stronger assertion, omit h^{-4} from the slack interval. Changing ξ_i to 1, for $i \leq R$, only removes points whose old slack is between 0 and $(\xi_i - 1)u_i$, which obey the same bound. Finally every full unit box has uniformly bounded mass when an arbitrary subset of its coordinates uses reciprocal-prime measure: multiply the factors in (3.12) for precisely that subset. This proves the mixed-measure assertion as well. $\square$

Lemma 3.7. *Uniformly for $Y \geq \exp(\exp 2)$ and $Z \geq 0$,*

$$\sum_{P^+(a) \leq Y} \frac{a^{1/\log Y}}{\varphi(a)} \ll \log Y, \quad \sum_{\substack{P^+(a) \leq Y \\ \log a > Z}} \frac{1}{\varphi(a)} \ll (\log Y) \exp(-Z/\log Y).$$

In particular, with $Y = \exp \exp(1.1 \tilde{b}_L)$,

$$\sum_{P^+(a) \leq Y} \frac{1}{\varphi(a)} \leq K_H, \quad \sum_{\substack{P^+(a) \leq Y \\ \log a > \exp(2 \tilde{b}_L)}} \frac{1}{\varphi(a)} \leq K_H \exp\{-\exp(cP\rho^{-P})\}. \quad (3.16)$$

Proof. Put $\sigma = 1/\log Y$. The Euler factor is

$$1 + \frac{p^\sigma}{(p-1)(1-p^{\sigma-1})},$$

whose logarithm is $p^{-1+\sigma} + O(p^{-2+2\sigma})$, uniformly in this range of σ . Partial summation gives $\sum_{p \leq Y} (p^{-1+\sigma} - p^{-1}) = O(1)$; hence the product is $O(\log Y)$. Rankin's inequality gives the tail bound. For the last assertion the exponent is

$$1.1\tilde{b}_L - \exp(0.9\tilde{b}_L),$$

and $\tilde{b}_L \asymp P\rho^{-P}$. $\square$

Combining Lemmas 3.6 and 3.7, the shell estimates remain valid after summing a smooth cofactor with weight $1/\varphi(a)$, at a cost of at most K_H . In particular their weighted mass is bounded by

$$CK_H G_m \left\{ H^{-2} + \sum_{h \geq H} (he^{-h/40} + h^3 \rho^h) \right\}. \quad (3.17)$$

The H^{-2} term is needed only for condition (iii).

3.5 Coverage of the values

Proof of Proposition 3.3. Take $\eta = 0.04$ in (3.7). By (3.2), its complementary bands imply the coarse bands in (3.10). Apply (3.8) as well. Both conclusions hold for all preimages outside the union of their exceptional sets.

We may also remove $v \leq x^{0.95}$ and $\Omega(v) > 5B$, at a cost $o(V(x))$. For the latter, the elementary reciprocal moment bound with $z = 1.9 < 2$ gives

$$\#\{v \leq x : \Omega(v) > 5B\} \leq z^{-5B} x \sum_{r \leq x} \frac{z^{\Omega(r)}}{r} \ll x \exp((z - 5 \log z)B) = o(x/\log x).$$

The Euler product bounds the reciprocal sum by $O((\log x)^z)$, and $z - 5 \log z < -1$. For every preimage, $\Omega(n) \leq \Omega(\varphi(n)) + 1 \leq 6B$. The p_1 band gives $\log p_1 \leq (\log x)^{0.61}$, so

$$\log p_0 \geq 0.95 \log x - 6B(\log x)^{0.61} > 0.9 \log x.$$

It remains to impose the size restriction on a . For any preimage failing it, fix $p_1, \dots, p_L, a$. The prime p_0 is distinct from the others and from the factors of a . Supermultiplicativity of φ gives

$$\varphi(p_0 \cdots p_L a) \geq (p_0 - 1) \varphi(a) \prod_{i=1}^L (p_i - 1).$$

The prime number theorem upper bound therefore gives at most

$$\frac{Cx}{\log x} \frac{1}{\varphi(a) \prod_{i=1}^L (p_i - 1)}$$

choices of p_0 ; if the interval is nonempty its upper endpoint is at least $x^{0.9}$. Sum the prefix by Lemma 3.6, and the cofactor by Lemma 3.7. This bounds all failing preimages, and therefore all values having at least one such preimage. It proves the exceptional-value assertion.

Finally $b_{i+1}/b_i = (1 - 1/h_i)\rho$. The bands and (3.2) imply that the largest double logarithm after index R is at most $0.62b_R$. There are $H - P \leq H$ such primes, proving the second line of (3.11). Its right-hand side is smaller than $\exp(0.89b_R) < \log p_R$ for large H , then large x . For the first line use the p_1 bound and $L \ll \log B$, while the a bound depends only on H . $\square$

3.6 Tuples and their discarded part

For $t = x$ or $t = x/c$, where $c > 1$ is fixed, keep all parameters formed from x . Let $\mathcal{T}(t)$ be the set of distinct tuples

$$\tau = (p_0, \dots, p_R, d)$$

admitting at least one basic witness with

$$d = \varphi(w), \quad w = p_{R+1} \cdots p_L a, \quad d \prod_{i=0}^R (p_i - 1) \leq t.$$

Each tuple represents a totient, because $w < p_R$ makes its tail coprime to its prefix. Several witnesses may give the same tuple; $\mathcal{T}(t)$ counts that tuple once.

We next remove tuples for which the comparison argument might fail. For each $0 \leq i \leq R$, put $h = h_i$ and

$$J_i = \lceil 30 \log h \rceil, \quad S_i = \exp \exp(b_i^{1/3}), \quad z_i = \exp \exp(0.7b_{i+J_i}),$$

and, for a basic witness, put

$$D_i = \varphi(p_{i+J_i+1} \cdots p_L a).$$

For large H these indices exist: $J_i < h_i/2$ and $i + J_i < L$. We call a tuple good if *every* basic witness giving it satisfies, for every $0 \leq i \leq R$,

$$\begin{aligned} \sum_{r=i+1}^L a_{r-i} u_r &\leq (1 - h_i^{-4}) u_i, \\ p_i, \dots, p_{i+J_i} &\text{ are } S_i\text{-normal}, \\ (p_i - 1) \cdots (p_{i+J_i} - 1) &\text{ is squarefree on primes exceeding } z_i, \\ \Omega(D_i) &\leq b_i/h_i^8. \end{aligned} \tag{3.18}$$

As before, the first condition uses $u_0 = B$. Denote the good tuples by $\mathcal{G}(t)$, and put $\mathcal{D}(t) = \mathcal{T}(t) \setminus \mathcal{G}(t)$.

Proposition 3.8. *For $t = x$ and $t = x/c$, with the same x -dependent parameters,*

$$|\mathcal{D}(t)| \leq \{\varepsilon_H + o_{x;H}(1)\} \frac{xG_m}{\log x}.$$

One may take $\varepsilon_H \ll K_H H^{-2}$. For all the witnesses in (3.18) we also have

$$P^+(D_i) < z_i, \quad S_i < z_i, \quad ch_i^{-20} \leq b_{i+J_i}/b_i \leq Ch_i^{-18}. \tag{3.19}$$

Proof. The last comparison follows from

$$\frac{b_{i+J_i}}{b_i} = (1 - J_i/h_i) \rho^{J_i}, \quad 18 < 30 \log(1/\rho) < 20.$$

It gives $S_i < z_i$. The next prime band has upper endpoint, in double logarithms, less than $0.7b_{i+J_i}$; all remaining primes, including those in a , are smaller. Taking totients cannot introduce a prime above this bound. This proves (3.19).

A bad tuple has at least one failing full witness. We bound the number of these witnesses; this is an upper bound for the number of bad tuples, regardless of witness multiplicity. The

prime-count upper bound for p_0 contributes $Cx/\log x$, with the product of reciprocal factors for all other variables. The smooth a -sum contributes at most K_H . Failure of the first condition in (3.18) is covered by Lemma 3.6, condition (iii). Its contribution is therefore $O(K_H H^{-2} x G_m / \log x)$.

For the other conditions fix i , write $h = h_i$, $b = b_i$, and $J = J_i$. If $i \geq 1$, the reciprocal sum for the common initial segment $p_1, \dots, p_{i-1}$ is $O(G_m)$, by (3.13). All of $p_i, \dots, p_L$ are at most $\exp \exp(1.2b)$. Their unrestricted reciprocal sums together cost $\exp(Ch^2)$, since there are at most h variables and $\log b = O(h)$. For $i = 0$, omit p_0 from these reciprocal sums and use the same estimate at $b = B$.

Partial summation of (3.9) gives

$$\sum_{\substack{p \leq \exp \exp(1.2b) \\ p \text{ not } S_i\text{-normal}}} \frac{1}{p-1} \ll (1+b)^6 e^{-b^{1/3}/6} \leq e^{-cb^{1/3}}.$$

For a failure involving p_0 , apply (3.9) directly to its counting interval; its upper endpoint has logarithm comparable with $\log x$. The normality failures thus contribute, relative to $x G_m / \log x$, at most

$$K_H \sum_{h \geq H} \exp(Ch^2 - cb(h)^{1/3}), \quad b(h) = \alpha h \rho^{-h}. \quad (3.20)$$

Polynomial factors from choosing an index are included in Ch^2 .

A square of a prime $q > z_i$ in the shifted-prime product either divides one shift twice or divides two different shifts. For $\ell = 1, 2$ and $Z \geq 2$, the elementary estimate

$$\sum_{\substack{p \leq Z \\ q^\ell | p-1}} \frac{1}{p-1} \leq \frac{1 + \log Z}{q^\ell}$$

follows by summing over all positive multiples of q^ℓ . If the shift is $p_0 - 1 \leq T$, its integer count is at most $\lfloor T/q^\ell \rfloor \leq T/q^\ell$. Relative to prime counting this loses at most $\log x = e^b$ when $i = 0$. Both types of square therefore cost q^{-2} , up to $\exp(Cb + Ch^2)$. Summing over $q > z_i$, their total relative contribution is at most

$$K_H \sum_{h \geq H} \exp\{Ch^2 + Cb(h) - \exp(cb(h)h^{-20})\}. \quad (3.21)$$

Finally $\Omega(\varphi(a)) \ll \log a \leq e^{\widetilde{2b}_L} = H^{o(1)}$, whereas b_i/h_i^8 is exponentially large in $h_i \geq H$. In the identity for D_i , overlap between a and the displayed primes can occur only at p_L . That overlap changes the sum of factor counts by $1 - \Omega(p_L - 1) \leq 0$. Failure of the last condition in (3.18) therefore forces some $i + J < j \leq L$ to satisfy $\Omega(p_j - 1) > b_i/(2h^9)$. The Euler product gives

$$\sum_{n \leq Z} \frac{(3/2)^{\Omega(n)}}{n} \ll (\log Z)^{3/2}.$$

Using the band bound for p_j , this exceptional reciprocal sum is at most

$$\exp(Cb_{i+J} - cb_i/h^9) \leq \exp(-c'b_i/h^9),$$

because $b_{i+J} \ll b_i h^{-18}$. The resulting relative contribution is at most

$$K_H \sum_{h \geq H} \exp(Ch^2 - cb(h)/h^9). \quad (3.22)$$

Each sum in (3.20)–(3.22) decreases faster than every inverse power of H , uniformly for α in its compact range. Together with the thin-slack bound, this proves the proposition. Replacing the endpoint x by x/c only decreases the witness sets, so the same bound holds with unchanged parameters. $\square$

The two cuts have now served different purposes. The cut at L keeps the smooth-cofactor cost at $K_H = H^{o(1)}$; the earlier cut at R supplies the summable H^{-2} loss and the much stronger savings in the remaining discards. Proposition 3.3 controls exceptional *values*, while Proposition 3.8 controls exceptional *tuples*. Neither yet asserts uniqueness of a representation. That is the purpose of the comparison and collision arguments that follow.

4 Comparing products of shifted primes

We next bound the number of ways that two ordered lists of primes can give the same product of their shifts. The largest prime factors of the shifts lie in separated intervals. We expose the factors one interval at a time, starting with the smallest interval. At each stage only two or three simultaneous prime conditions are needed. This keeps the constants uniform when the number of intervals grows.

The layer decomposition and the comparison of dual factorizations go back to Maier and Pomerance [10, Section 3] and were developed further in Ford’s proof of Lemma 5.1 [7]. We prove the version needed here, including its dependence on the number of layers. In particular, we assume squarefreeness of the whole product above the last cutoff; squarefreeness of each individual shift would not suffice for our factor-allocation estimate.

4.1 A uniform sieve for two or three forms

Lemma 4.1. *There are absolute constants C and y_0 with the following property. Let $y \geq y_0$, put $B_y = \log_2 y$, and let $1 \leq A, A' \leq y$ be integers. Consider either the two forms*

$$n, \quad An + 1,$$

or the three forms

$$n, \quad An + 1, \quad A'n + 1, \quad A \neq A'.$$

Write $k = 2$ or 3 , respectively, and let $\mathcal{P}$ be the set of positive integers at which all the indicated forms are prime. Then, uniformly for $X \geq 3$,

$$\#(\mathcal{P} \cap [1, X]) \leq C \frac{XB_y^k}{(\log X)^k}. \quad (4.1)$$

Moreover, uniformly for $3 \leq U \leq Y$,

$$\sum_{\substack{n \in \mathcal{P} \\ U \leq n \leq Y}} \frac{1}{n} \leq C \frac{B_y^k}{(\log U)^{k-1}}. \quad (4.2)$$

The coefficients need not be bounded in terms of X or U .

Proof. For primitive, distinct, admissible integral linear forms $a_j n + b_j$ with positive leading coefficients and nonzero determinant Δ as defined below, Ford's uniform upper sieve [8, Theorem 2.5] gives

$$\#\{1 \leq n \leq X : a_j n + b_j \text{ prime for every } j\} \ll_k \frac{X}{(\log X)^k} \left(\frac{\Delta}{\varphi(\Delta)} \right)^k, \quad (4.3)$$

$$\Delta = \left| \prod_j a_j \prod_{j < l} (a_j b_l - a_l b_j) \right|.$$

Here admissibility means that no prime divides the product of the forms for every integer input; the implied constant depends only on k , including when the coefficients vary. For our forms, primitivity is automatic, and the determinants are A and $AA'(A - A')$, up to sign. They are nonzero and have absolute value at most y^3 .

If our forms are inadmissible, an offending prime is at most k , since each primitive linear form has at most one root modulo that prime. At every simultaneous-prime input one of the forms must equal this prime. Each such equality has at most one solution, so there are at most k inputs. Their contribution is absorbed in (4.1).

For admissible forms, the elementary estimate

$$\frac{d}{\varphi(d)} \ll \log_2(d + 10) \quad (d \geq 1)$$

and (4.3) prove (4.1). To recall its uniformity, split the product over prime divisors of d at $\log(d + 10)$. Mertens' product estimate bounds the first part by $O(\log_2(d + 10))$. For the remaining part, use $\sum_{p|d} \log p \leq \log d$ to bound the sum of reciprocal prime divisors, giving a bounded factor. Since k takes only the values 2 and 3, all constants are absolute.

Finally, partial summation of (4.1), with the coefficients fixed, gives

$$\sum_{\substack{n \in \mathcal{P} \\ U \leq n \leq Y}} \frac{1}{n} \ll B_y^k \left(\frac{1}{(\log Y)^k} + \int_U^Y \frac{dt}{t(\log t)^k} \right) \ll \frac{B_y^k}{(\log U)^{k-1}}.$$

This also includes a possible prime at the lower endpoint. □

4.2 The comparison estimate

For a positive integer n and a real number Z , its part supported above Z means $\prod_{p^a \parallel n, p > Z} p^a$. We use S -normality as defined in Definition 3.1.

The key term in the bound below is the exponent of $\log y$. In its application, a strict simplex inequality makes the weighted cutoff sum plus its error smaller than 1 by a controlled amount. The exponent is consequently smaller than -1 ; that extra saving absorbs the other factors, even after recovering the tuple data hidden in the smooth remainder. We need an estimate for ordered pairs with multiplicity: a bound only on the number of values admitting a collision would not control how many tuples lie over those values.

Proposition 4.2. *There are absolute constants C and y_0 such that the following holds. Let $y \geq y_0$, put $B_y = \log_2 y$, and let b be an integer with $1 \leq b \leq B_y$. Suppose that real cutoffs satisfy*

$$Y_0 = y, \quad e^e \leq S \leq Y_b, \quad Y_{j+1} < U_j < Y_j \quad (0 \leq j < b).$$

Put

$$\nu_j = \frac{\log_2 Y_j}{B_y} \quad (0 \leq j \leq b), \quad \mu_j = \frac{\log_2 U_j}{B_y} \quad (0 \leq j < b), \quad \delta = \sqrt{\frac{\log_2 S}{B_y}}.$$

Fix a positive integer D with $P^+(D) \leq Y_b$ and a real $r \geq 1$. Let N count the ordered tuples

$$(p_0, \dots, p_{b-1}, q_0, \dots, q_{b-1}, D')$$

in which D' is a positive integer and the following conditions hold:

(i) The p_j, q_j are S -normal primes, $p_j \neq q_j$, and

$$U_j \leq P^+(p_j - 1), P^+(q_j - 1) \leq Y_j \quad (0 \leq j < b).$$

(ii) $P^+(D') \leq Y_b$ and

$$D \prod_{j=0}^{b-1} (p_j - 1) = D' \prod_{j=0}^{b-1} (q_j - 1) \leq \frac{y}{r}.$$

(iii) The common product in (ii) is squarefree on primes greater than Y_b .

(iv) The part of $p_0 - 1$ supported above Y_1 exceeds $\sqrt{y}$.

Then

$$N \leq \frac{y}{rD} (CB_y^6)^b (b+1)^{\Omega(D)} (\log Y_b)^{1+3b \log(b(b+1))} \times (\log y)^{-2+\sum_{j=1}^{b-1} a_j \nu_j + E_b}, \quad (4.4)$$

where

$$E_b = 2 \sum_{j=1}^{b-1} (\nu_j - \mu_j) + \delta \sum_{i=2}^b (i \log i + i). \quad (4.5)$$

Here a_j is as defined in (2.1). Empty sums are zero; in particular, when $b = 1$ the cutoff in (iv) is $Y_1 = Y_b$.

Proof. We first describe the layers and the weighted summation that combines them. We then count the top layer, estimate a general lower layer, and finally sum over the small prime factors.

The layer configurations. For $0 \leq i \leq b$ and $0 \leq j < b$, let

$$s_{i,j} = \prod_{\substack{p^a \parallel p_j - 1 \\ p \leq Y_i}} p^a, \quad s'_{i,j} = \prod_{\substack{p^a \parallel q_j - 1 \\ p \leq Y_i}} p^a.$$

Restricting the equality in condition (ii) to primes at most Y_i gives the common product

$$s_i^\# = D \prod_{j=0}^{b-1} s_{i,j} = D' \prod_{j=0}^{b-1} s'_{i,j}.$$

For $1 \leq i \leq b$, put

$$t_{i,j} = \frac{s_{i-1,j}}{s_{i,j}}, \quad t'_{i,j} = \frac{s'_{i-1,j}}{s'_{i,j}}, \quad t_i^\# = \frac{s_{i-1}^\#}{s_i^\#}.$$

Each $t_i^\#$ is squarefree and supported on $(Y_i, Y_{i-1}]$. For $j \geq i$ the factors $t_{i,j}$ and $t'_{i,j}$ equal 1, so

$$t_i^\# = \prod_{j=0}^{i-1} t_{i,j} = \prod_{j=0}^{i-1} t'_{i,j}. \quad (4.6)$$

Thus a layer consists of two allocations of the same squarefree integer among i labeled slots.

Let $\mathcal{C}_i$ be the set of configurations

$$\sigma_i = (D', s_{i,0}, \dots, s_{i,b-1}, s'_{i,0}, \dots, s'_{i,b-1})$$

arising from tuples counted by N . Restriction of prime factors maps $\mathcal{C}_{i-1}$ into $\mathcal{C}_i$; the two allocations in (4.6) determine each extension uniquely. Reconstructing a tuple from its small prime factors runs from $\mathcal{C}_b$ back to $\mathcal{C}_0$; the weighted count below sums in the opposite direction, starting with the top layer. The original tuples are in bijection with $\mathcal{C}_0$, since $p_j = s_{0,j} + 1$ and $q_j = s'_{0,j} + 1$. For $i \geq 2$ we will bound, uniformly in σ_i , the reciprocal extension sum

$$\sum_{\substack{\sigma_{i-1} \in \mathcal{C}_{i-1} \\ \sigma_{i-1} \text{ restricts to } \sigma_i}} \frac{1}{t_i^\#}.$$

This is the appropriate weight because $1/s_{i-1}^\# = 1/(s_i^\# t_i^\#)$.

Every configuration considered here extends to a genuine solution. In particular, its partial common product is at most y/r , each shift is at most y , and every fixed or conditional coefficient used below is at most y . When enlarging a sum for an upper bound, we retain any necessary restriction on a conditional coefficient until the corresponding sieve estimate has been applied.

The top layer. Only the zeroth shift can have a prime factor greater than Y_1 . Consequently $t_1^\#$ is the same part of $p_0 - 1$ and $q_0 - 1$, and $t_1^\# > \sqrt{y}$. Normality, applied with endpoint $p_0 - 1$, gives

$$\Omega(p_0 - 1) \leq B_y + \log_2 S + \sqrt{B_y \log_2 S} \leq 3B_y.$$

Write $t_1^\# = t'Q$, with $Q = P^+(t_1^\#)$. The weaker bound $\Omega(t_1^\#) \leq 4B_y$ already gives

$$Q \geq y^{1/(8B_y)}.$$

After σ_1 and t' are fixed, the completed primes have the forms $AQ + 1, A'Q + 1$. Their inequality implies $A \neq A'$, which is exactly the nonzero determinant condition for the three-form sieve. With

$$X = \frac{y}{rs_1^\# t'},$$

a nonempty extension has $X \geq Q \geq y^{1/(8B_y)}$. Lemma 4.1 therefore bounds the number of Q by

$$\ll \frac{XB_y^3}{(\log X)^3} \ll \frac{y}{rs_1^\# t'} \frac{B_y^6}{(\log y)^3}.$$

The integer t' is squarefree and supported on $(Y_1, y]$. Mertens' estimate gives

$$\sum_{t'} \frac{1}{t'} \leq \prod_{Y_1 < p \leq y} \left(1 + \frac{1}{p}\right) \ll \frac{\log y}{\log Y_1}.$$

It follows that, for each σ_1 ,

$$\#\{\sigma_0 \text{ restricting to } \sigma_1\} \leq CB_y^6 \frac{y}{rs_1^\#} (\log y)^{-2-\nu_1}. \quad (4.7)$$

The prime conditions in an intermediate layer. Fix $2 \leq i \leq b$ and σ_i . The last slot, of index $i - 1$, is complete after this layer. Let Q_1, Q_2 be its largest prime factors on the two sides; both lie in $[U_{i-1}, Y_{i-1}]$.

Suppose first that $Q_1 = Q_2 = Q$. Remove the unique occurrence of Q from $t_i^\#$, and call the result t . For fixed allocations of t , the completed primes are $AQ + 1, A'Q + 1$ with $A \neq A'$. Lemma 4.1 gives

$$\sum_Q \frac{1}{Q} \ll \frac{B_y^3}{(\log U_{i-1})^2}. \quad (4.8)$$

Now suppose $Q_1 \neq Q_2$ and remove both, obtaining t . In addition to its two allocations, fix the position of Q_1 on the second side and of Q_2 on the first side. There are at most i^2 choices. If neither prime lies in the opposite last slot, the two completed prime conditions are $AQ_1 + 1$ and $A'Q_2 + 1$; two applications of (4.2) with $k = 2$ give

$$\sum_{Q_1, Q_2} \frac{1}{Q_1 Q_2} \ll \frac{B_y^4}{(\log U_{i-1})^2}. \quad (4.9)$$

If Q_2 lies in the last slot on the first side, then $Q_2 < Q_1$. The prime Q_1 cannot lie in the last slot on the second side, whose largest prime is Q_2 . The completed prime conditions are now $AQ_2 Q_1 + 1$ and $A'Q_2 + 1$. First sum over Q_1 conditional on Q_2 , using the forms $Q_1, AQ_2 Q_1 + 1$. Feasibility ensures $AQ_2 \leq y$, so the inner reciprocal sum is uniformly $O(B_y^2 / \log U_{i-1})$. Only after this estimate do we enlarge the range of Q_2 and apply the two-form bound to $Q_2, A'Q_2 + 1$. This proves (4.9) again. The reversed incidence is treated by reversing the order of summation. Both incidences cannot occur simultaneously, as they would imply both $Q_1 < Q_2$ and $Q_2 < Q_1$.

Counting the allocations. Put $\Delta_i = \nu_{i-1} - \nu_i$. Normality bounds the number of prime factors of an individual shift in $(Y_i, Y_{i-1}]$ by $(\Delta_i + \delta)B_y$. Indeed, if the shift ends before Y_{i-1} , apply normality only up to its endpoint; if it ends before Y_i , the count is zero. In either case the error is at most $\sqrt{\log_2 S B_y} = \delta B_y$. Only i shifts contribute to this layer, and removing the distinguished primes decreases the count. Thus

$$\Omega(t) \leq I_i := i(\Delta_i + \delta)B_y.$$

Since t is squarefree, each of its prime factors chooses one of i slots on each side, giving at most $i^{2\Omega(t)}$ dual allocations. Write

$$M_i = \sum_{Y_i < p \leq Y_{i-1}} \frac{1}{p}.$$

Mertens' estimate gives $M_i \leq \Delta_i B_y + O(1)$, uniformly in the cutoffs. Since $\delta B_y \geq \sqrt{B_y}$, increasing the absolute threshold y_0 ensures $M_i \leq (\Delta_i + \delta)B_y$. Hence

$$\begin{aligned} \sum_t \frac{i^{2\Omega(t)}}{t} &\leq \sum_{n \leq I_i} \frac{(i^2 M_i)^n}{n!} \leq i^{I_i} \exp(i M_i) \\ &\leq \exp((i \log i + i)(\Delta_i + \delta)B_y). \end{aligned} \quad (4.10)$$

The middle inequality follows by writing $i^{2n} = i^n i^n$, using $i^n \leq i^{I_i}$, and extending the remaining exponential series. It is valid without any lower bound on Δ_i / δ .

Combining (4.8), (4.9), and (4.10), and using $i \leq b \leq B_y$ to absorb the i^2 position choices, gives

$$\sum_{\substack{\sigma_{i-1} \text{ restricting} \\ \text{to } \sigma_i}} \frac{1}{t_i^\#} \leq C B_y^6 (\log y)^{-2\mu_{i-1} + (i \log i + i)(\nu_{i-1} - \nu_i + \delta)}. \quad (4.11)$$

The squarefree hypothesis was used both to remove a single occurrence of each distinguished prime and to obtain the factorial denominator in (4.10).

Multiplying the layer estimates. Starting from (4.7), sum over $\mathcal{C}_1$. For each subsequent layer use the exact weighted identity

$$\sum_{\sigma_{i-1} \in \mathcal{C}_{i-1}} \frac{1}{s_{i-1}^\#} = \sum_{\sigma_i \in \mathcal{C}_i} \frac{1}{s_i^\#} \sum_{\sigma_{i-1} \text{ restricting to } \sigma_i} \frac{1}{t_i^\#}.$$

All bounds in (4.11) are uniform in the lower configuration. Their constants therefore multiply to at most $C^b B_y^{6b}$. If $c_i = i \log i + i$, so that $c_1 = 1$, the power of $\log y$ accumulated in this procedure is

$$\begin{aligned} & -2 - \nu_1 + \sum_{i=2}^b \{-2\mu_{i-1} + c_i(\nu_{i-1} - \nu_i + \delta)\} \\ & = -2 + \sum_{j=1}^{b-1} a_j \nu_j + 2 \sum_{j=1}^{b-1} (\nu_j - \mu_j) + \delta \sum_{i=2}^b c_i - c_b \nu_b. \end{aligned}$$

Here $c_{j+1} - c_j - 2 = a_j$, including $j = 1$. The identity also holds for $b = 1$. Discard the nonpositive final term $-c_b \nu_b$. It remains to bound $\sum_{\sigma_b \in \mathcal{C}_b} 1/s_b^\#$.

The smooth parts and repeated factors. Write $s = s_b^\# / D$ and $L_b = \log_2 Y_b$. For a fixed s , its allocation among the b left slots has at most $b^{\Omega(s)}$ possibilities. The allocation of sD among the b right slots and the additional slot D' has at most $(b+1)^{\Omega(sD)}$ possibilities. These bounds allow prime powers: assigning labeled copies of a repeated prime to slots overcounts every possible distribution of its exponent.

Each left shift has largest prime factor above Y_b . Its number of prime factors at most Y_b is bounded by normality as follows:

$$\Omega(p_j - 1, 1, Y_b) \leq 2 \log_2 S + (L_b - \log_2 S) + \sqrt{L_b \log_2 S} \leq 3L_b.$$

If $S = Y_b$, the first normality inequality alone gives the same bound. Thus $\Omega(s) \leq 3bL_b$. The ordinary smooth-number Euler product, including all powers, now gives

$$\begin{aligned} \sum_{\sigma_b \in \mathcal{C}_b} \frac{1}{s_b^\#} & \leq \frac{(b+1)^{\Omega(D)}}{D} [b(b+1)]^{3bL_b} \sum_{P^+(s) \leq Y_b} \frac{1}{s} \\ & = \frac{(b+1)^{\Omega(D)}}{D} [b(b+1)]^{3bL_b} \prod_{p \leq Y_b} \left(1 - \frac{1}{p}\right)^{-1} \\ & \ll \frac{(b+1)^{\Omega(D)}}{D} (\log Y_b)^{1+3b \log(b(b+1))}. \end{aligned}$$

No squarefreeness is assumed below Y_b . Combining this estimate with the layer bounds proves (4.4), after increasing the absolute constant C . $\square$

Remark 4.3. The explicit error in (4.5) satisfies

$$E_b = 2 \sum_{j=1}^{b-1} (\nu_j - \mu_j) + O(\delta b^2 \log(2b)),$$

and the smooth-factor exponent is $O(b \log(2b))$. Thus all dependence on a growing b is displayed in (4.4). No restriction such as $r \leq y^{1/10}$ or $D \leq y^{1/100}$ is needed: the large top part supplies the lower bound for the actual sieve endpoint.

5 Uniqueness of the continuous prefix

We now pass from representations to distinct totient values. We retain the tuple sets $\mathcal{T}(t)$, $\mathcal{G}(t)$ and $\mathcal{D}(t)$ of Section 3: the first consists of the distinct tuples $(p_0, \dots, p_R, d)$ admitting a basic witness, the second consists of those for which every basic witness satisfies (3.18), and $\mathcal{D}(t) = \mathcal{T}(t) \setminus \mathcal{G}(t)$. Their value map is

$$F(p_0, \dots, p_R, d) = d \prod_{j=0}^R (p_j - 1).$$

The tail preimages of d are not part of a tuple. This distinction is essential: we shall prove that almost every value has one such tuple, while allowing several tail preimages.

5.1 Counting pairs with the same value

The alignment of largest factors and cancellation follow the argument around (5.34)–(5.37) in Ford’s Section 5 [7]. Here we count all ordered pairs of distinct tuples, including the data inside both residual factors. We first cancel the common prefix and align the large factors in the remaining shifts. After applying the comparison estimate, we recover the tuple coordinates inside the residual factors and then restore the common prefix.

Proposition 5.1 (Prefix collisions). *Fix $c > 1$, and form all parameters from x , with H fixed before x tends to infinity. For $t = x$ or $t = x/c$, let*

$$K(t) = \#\{(\tau, \sigma) \in \mathcal{G}(t)^2 : \tau \neq \sigma, F(\tau) = F(\sigma)\}.$$

There is a quantity $\epsilon_H \rightarrow 0$, independent of the phase, such that

$$K(t) \leq (\epsilon_H + o_{x,H}(1)) \frac{x G_m}{\log x}.$$

The pairs counted here are ordered pairs of distinct tuples.

Proof. Choose one basic witness for each tuple under consideration. All choices satisfy (3.18), by the definition of $\mathcal{G}(t)$. If two distinct tuples have the same value, some displayed prime differs: equality of all the primes $p_0, \dots, p_R$ would also force equality of d . Let i be the first unequal prime index, and write q_j for the primes in the other witness. Put

$$h = m - i, \quad J = \lceil 30 \log h \rceil, \quad z = z_i = \exp \exp(0.7 b_{i+J}), \quad S = S_i = \exp \exp(b_i^{1/3}).$$

Thus $h \geq H$. The residual integers in the good conditions are

$$D_i = \varphi(p_{i+J+1} \cdots p_L a), \quad D'_i = \varphi(q_{i+J+1} \cdots q_L a').$$

Canceling the common prefix leaves the identity

$$v^{(i)} = D_i \prod_{j=0}^J (p_{i+j} - 1) = D'_i \prod_{j=0}^J (q_{i+j} - 1). \quad (5.1)$$

We will apply Proposition 4.2 to this identity. First we verify its size and interval hypotheses; then we count all ways in which its solutions can give tuple pairs.

Sizes of the suffix and its layers. For sufficiently large H , uniformly for $h \geq H$,

$$b_i \asymp h\rho^{-h}, \quad \log b_i = O(h), \quad c_1 h^{-20} \leq \frac{b_{i+J}}{b_i} = (1 - J/h)\rho^J \leq C_1 h^{-18}. \quad (5.2)$$

Indeed 30λ lies strictly between 18 and 20; the ceiling in J changes only constant factors. In particular $J < h/2$ and $i + J < L$. The constants in this proof are independent of H, x, i and the phase, once H and then x are sufficiently large.

For $i = 0$ set $y = x$. For $i \geq 1$ place $v^{(i)}$ in a dyadic interval $(y/2, y]$. The coarse bands in (3.10) imply that the logarithm of the preimage suffix after p_i is at most

$$h \exp(0.62b_i) + \exp(2\tilde{b}_L).$$

To see the first term, use $1.1\tilde{b}_{i+1} < 0.62b_i$ for large x , and bound the number of displayed factors by h . The bound for a gives the second term. These quantities are negligible compared with $\log p_i \geq \exp(0.88b_i)$. Consequently, writing $B_y = \log_2 y$,

$$p_i, q_i \geq y^{0.9}, \quad 0.88b_i \leq B_y \leq 1.12b_i, \quad u_i \leq B_y + O(1). \quad (5.3)$$

For $i = 0$ the first inequality is part of the basic witness conditions, and $B_y = b_0 = u_0 = B$ by convention.

Our target is a bound of the form

$$\frac{y}{\log y} \exp(-\kappa B_y / h^4)$$

for the suffix pairs, with an absolute $\kappa > 0$. The strict simplex slack will supply the exponential saving. Since $B_y \asymp b_i$, this saving can absorb the $\exp(O(b_i \log h / h^8))$ cost of recovering tuple coordinates inside D_i and D'_i .

Let

$$\delta = \sqrt{\log_2 S / B_y} \asymp b_i^{-1/3}.$$

It follows from (5.2) that $S < z$ and $J\delta = o(h^{-20})$. Normality gives $\Omega(p_{i+j} - 1), \Omega(q_{i+j} - 1) \leq 4B_y$ for $0 \leq j \leq J$. For example, split the prime factors at S , apply the two normality conditions, and use $p_{i+j} - 1 \leq v^{(i)} \leq y$. If

$$A_j = P^+(p_{i+j} - 1), \quad A'_j = P^+(q_{i+j} - 1),$$

then

$$\log_2 A_j = u_{i+j} + O(\log b_i), \quad \log_2 A'_j = \log_2 q_{i+j} + O(\log b_i). \quad (5.4)$$

This follows from $(p_{i+j} - 1) \leq A_j^{\Omega(p_{i+j}-1)}$ and the reverse bound $A_j \leq p_{i+j} - 1$. When $i = j = 0$, the convention $u_0 = B$ causes only an $O(1)$ difference from $\log_2 p_0$, since $x^{0.9} \leq p_0 \leq x + 1$.

The bands in (5.4) are disjoint and decrease with j . Their smallest separation is a constant multiple of $b_i h^{-20}$, which dominates the error $O(\log b_i)$. All the A_j, A'_j exceed z , whereas

$$P^+(D_i), P^+(D'_i) < z.$$

For the latter assertion, every prime in the residual preimage is at most p_{i+J+1} or q_{i+J+1} , and $1.1\tilde{b}_{i+J+1} < 0.7b_{i+J}$; taking a totient cannot increase the largest prime factor. Repetitions of the last prime in a or a' do not affect this conclusion.

Aligning the largest factors on the two sides. We claim that

$$|\log_2 A_j - \log_2 A'_j| \leq (2j+1)\delta B_y \quad (0 \leq j \leq J). \quad (5.5)$$

Suppose, for example, that $A_j > A'_j$, and put $e = (\log_2 A_j - \log_2 A'_j)/B_y$. On the first side of (5.1), each of its first $j+1$ shifts contributes at least $(e-\delta)B_y$ prime factors, with multiplicity, to $(A'_j, A_j]$. On the other side only its first j shifts can contribute, each at most $(e+\delta)B_y$. The later shifts have smaller largest factors, and both residual integers are z -smooth. The lower endpoint exceeds S , so normality applies. Equality of the two sides therefore gives

$$(j+1)(e-\delta) \leq j(e+\delta),$$

which proves (5.5). Interchanging the sides handles the other order.

Cancel any equal primes among the two lists in (5.1), and let r be the product of their shifts. The disjoint prime bands show that equality is possible only at the same original index. The index i survives, since $p_i \neq q_i$. Let $b \leq J+1 \leq B_y$ be the length of the remaining lists. Their corresponding primes are unequal, as required by Proposition 4.2.

Set $Y_b = z$, $Y_0 = y$ and $\log_2 U_0 = 0.8B_y$. At each remaining positive index, choose an interval on a δ -grid in normalized double logarithms which contains the corresponding pair from (5.5). Its width is $O(J\delta)$. The intervals remain strictly separated, because $J\delta = o(h^{-20})$. Each lies below height 0.8, by (5.3) and the upper bound $u_{i+1} \leq 0.62b_i$. At index zero, (5.3) and (5.4) give largest factors above height 0.9. Thus all the ordered-cutoff hypotheses of Proposition 4.2 hold. If $b = 1$, there are no positive indices to grid and $Y_1 = z$.

The part of $p_i - 1$ supported on primes at most Y_1 has logarithm at most $4B_y \log Y_1 \leq 4B_y \exp(0.8B_y)$. Since $p_i \geq y^{0.9}$, the complementary part exceeds $\sqrt{y}$ for large H . The whole common product is squarefree above Y_b , by the good conditions; canceling the equal shifts preserves this property. This verifies the last two hypotheses of Proposition 4.2.

Apply that proposition to the remaining lists with $D = D_i$, $D' = D'_i$ and the factor r just canceled; their common product is at most y/r . Write $\nu_k = \log_2 Y_k/B_y$ and $\mu_k = \log_2 U_k/B_y$ for the normalized interval endpoints.

Write $j_1 < \dots < j_{b-1}$ for the surviving positive original indices. Since $j_k \geq k$ and $a_k \leq a_{j_k}$, the grid intervals and the strict slack condition in (3.18) give

$$\begin{aligned} \sum_{k=1}^{b-1} a_k \nu_k &\leq \frac{1}{B_y} \sum_{j=1}^J a_j u_{i+j} + O(J^2 \log(2J)\delta) \\ &\leq 1 - h^{-4} + O(B_y^{-1} + J^2 \log(2J)\delta). \end{aligned} \quad (5.6)$$

Here a_j is increasing and $\sum_{j \leq J} a_j = O(J \log(2J))$. The grid widths and the explicit δ term in (4.4) contribute another $O(J^2 \log(2J)\delta) = o(h^{-4})$ to its exponent of $\log y$. Thus, for some absolute

$\kappa > 0$, its logarithmic factor is at most

$$(\log y)^{-1-\kappa h^{-4}} = \frac{1}{\log y} \exp(-\kappa B_y/h^4).$$

It remains to show that the other factors and the recovery of tuple coordinates use only a smaller part of this saving.

Recovering tuples and summing the remaining choices. The comparison estimate counts its displayed primes and its residual integer D'_i . We must also account for the tuple data forgotten inside D_i, D'_i . If $R \leq i + J$, all displayed tuple primes are already determined. Their tail totient is

$$d = D_i \prod_{j=R+1}^{i+J} (p_j - 1),$$

and similarly on the other side, with an empty product when $R = i + J$. If $R > i + J$, the missing tuple data give an ordered factorization

$$D_i = d \prod_{j=i+J+1}^R (p_j - 1). \quad (5.7)$$

There are at most $(h+1)^{\Omega(D_i)}$ such factorizations. Indeed, for any integer n , assigning its $\Omega(n)$ prime occurrences to k labeled slots bounds the number of ordered multiplicative factorizations into k factors by $k^{\Omega(n)}$. Once a factor $p_j - 1$ is fixed, p_j is fixed as well. This argument also covers repeated prime factors. Consequently the two missing tails cost at most

$$(h+1)^{\Omega(D_i)+\Omega(D'_i)} \leq \exp\{2b_i \log(h+1)/h^8\}. \quad (5.8)$$

The factorizations in (5.7) are exact: the primes up to R are distinct and the remaining preimage tail is smaller than p_R , by (3.11). All repetitions inside that tail are kept in its totient d .

We now sum the comparison bound over the canceled primes, the grids, and D_i . The canceled index sets contribute at most 2^{J+1} . Each canceled prime has reciprocal shift sum $O(b_i)$ by Mertens' theorem. The number of grid choices is at most $(C/\delta)^{2(J+1)}$. Thus their total cost, including the factors $(CB_y^6)^b$ in (4.4), is $\exp(O(h^2))$. For the residual integer, the finite Euler product gives

$$\sum_{P^+(D) \leq z} \frac{1}{D} \ll \log z.$$

Before enlarging this sum we use the good-condition bounds $\Omega(D_i), \Omega(D'_i) \leq b_i/h^8$ in both (5.8) and the factor $(b+1)^{\Omega(D_i)}$ of the comparison estimate.

The logarithm of all the remaining costs, apart from the factor $y/(\log y)$ and the saving from (5.6), is at most

$$O(h^2) + O\left(\frac{b_i \log h}{h^8}\right) + O\left(\frac{b_i \log h \log \log(3h)}{h^{18}}\right) = o\left(\frac{b_i}{h^4}\right). \quad (5.9)$$

The last term uses $\log_2 z \ll b_i h^{-18}$ to bound the factor $(\log Y_b)^{1+3b \log(b(b+1))}$ in (4.4), as well as the sum over D_i . All little-oh estimates here are uniform for $h \geq H$ as $H \rightarrow \infty$, because $b_i \asymp h \rho^{-h}$. We have therefore obtained, for some absolute $c_2 > 0$,

$$\#\{\text{ordered suffix tuple pairs with } y/2 < v^{(i)} \leq y\} \ll \frac{y}{\log y} \exp\{-c_2 B_y/h^4\}. \quad (5.10)$$

For $i = 0$ the same expression bounds all the pairs with first unequal index zero, using $y = x$. Choosing one witness for each tuple has not introduced multiplicity into the desired count: comparison data and the factorizations just counted determine each ordered pair of tuples.

Summing suffixes and restoring the common prefix. For $i \geq 1$ the required sum over suffix pairs is weighted by $1/v^{(i)}$. A dyadic interval in (5.10) contributes at most

$$\frac{C}{\log y} \exp\{-c_2 \log_2 y/h^4\}.$$

Writing the dyadic endpoints as $y = 2^n$, and comparing the sum with an integral, gives

$$\begin{aligned} \sum_{\text{suffix pairs}} \frac{1}{v^{(i)}} &\ll \sum_{\log(n \log 2) \geq 0.88b_i - O(1)} \frac{\exp\{-c_2 \log(n \log 2)/h^4\}}{n \log 2} \\ &\ll \int_{0.88b_i - O(1)}^{\infty} \exp(-c_2 u/h^4) du \ll h^4 \exp(-c_3 b_i/h^4). \end{aligned} \quad (5.11)$$

In this change of variable, $u = \log(n \log 2)$, the factor $1/\log y$ exactly compensates for the density of dyadic endpoints. In particular, no factor of order $\exp(b_i)$ is lost.

For fixed suffix data and common primes $p_1, \dots, p_{i-1}$, the number of possible p_0 is at most

$$\frac{Cx}{\log x} \frac{1}{v^{(i)} \prod_{j=1}^{i-1} (p_j - 1)}.$$

Indeed the upper endpoint for p_0 is $1 + x/(v^{(i)} \prod_{j=1}^{i-1} (p_j - 1))$; if the range is nonempty, its restriction $p_0 \geq x^{0.9}$ makes its logarithm comparable to $\log x$, so the prime number theorem gives this upper bound. The projected-prefix estimate (3.13) and (5.11) now show that the contribution for this i is at most

$$\frac{CxG_m}{\log x} h^4 \exp(-c_3 b_i/h^4).$$

This also applies to the empty projected prefix when $i = 1$. Summing over $h \geq H$ yields the convergent tail

$$C \sum_{h \geq H} h^4 \exp\{-c_4 \rho^{-h}/h^3\} \rightarrow 0.$$

For $i = 0$, (5.10) gives $O((x/\log x) \exp\{-c_2 B/m^4\})$, which is $o(xG_m/\log x)$. This proves the proposition, uniformly for both endpoints t . $\square$

5.2 From tuples to values and least preimages

The following elementary lemma keeps track of exceptional tuples as well as exceptional values. This is needed because a small set of values could otherwise carry many representations.

Lemma 5.2 (Finite-map counting). *Let T and W be finite sets, let $F : T \rightarrow W$, and write $T = G \sqcup D$. Suppose that every element of $W \setminus E$ is in $F(T)$, where $E \subset W$. For $w \in W$ put $r_w = |G \cap F^{-1}(w)|$, and let*

$$K = \sum_{w \in W} r_w(r_w - 1).$$

Then

$$-|E| \leq |T| - |W| \leq |D| + K/2.$$

Moreover, with $E^* = E \cup F(D) \cup \{w : r_w \geq 2\}$, one has

$$|E^*| \leq |E| + |D| + K/2, \quad |F^{-1}(E^*)| \leq |E| + 2|D| + K.$$

The restriction of F to $T \setminus F^{-1}(E^*)$ is a bijection onto $W \setminus E^*$.

Proof. Coverage gives the lower bound. For every positive integer r , $r - 1 \leq r(r - 1)/2$, and hence

$$|G| - |F(G)| \leq K/2.$$

Adding the at most $|D|$ remaining elements proves the upper bound. There are at most $K/2$ fibers with $r_w \geq 2$, which proves the bound for $|E^*|$. Also

$$\sum_{w \in E^*} r_w \leq |E^*| + K/2,$$

by $r \leq 1 + r(r - 1)/2$ for integers $r \geq 0$. Adding the elements of D proves the second bound. Outside E^* coverage, absence of discarded elements, and $r_w \leq 1$ give exactly one preimage under F . $\square$

Proposition 5.3 (A common prefix for all preimages). *For $t = x$ or $t = x/c$, with the same parameters formed from x ,*

$$||\mathcal{T}(t)| - V(t)| \leq (\epsilon_H + o_{x;H}(1)) \frac{xG_m}{\log x}, \quad \epsilon_H \longrightarrow 0. \quad (5.12)$$

There is a set $E^(t) \subset \mathcal{V} \cap [1, t]$ for which both $|E^*(t)|$ and the number of tuples in $\mathcal{T}(t)$ with values in $E^*(t)$ are at most $(\epsilon_H + o_{x;H}(1))xG_m/\log x$, after enlarging ϵ_H . For every $v \in (\mathcal{V} \cap [1, t]) \setminus E^*(t)$ there is a unique tuple $(p_0, \dots, p_R, d) \in \mathcal{T}(t)$ of value v , and every preimage of v has this prefix and a remaining factor of totient d . In particular,*

$$\ell(v) = p_0 \cdots p_R \ell(d). \quad (5.13)$$

Proof. Apply Lemma 5.2 with $T = \mathcal{T}(t)$, $G = \mathcal{G}(t)$, $D = \mathcal{D}(t)$ and $W = \mathcal{V} \cap [1, t]$. Let E be the exceptional values in Proposition 3.3, restricted to $[1, t]$. Every preimage of each $v \notin E$ is basic, so it induces an element of T . The size of E is negligible on the scale $xG_m/\log x$, by (3.4); Proposition 3.8 gives the same bound for $|D|$, and Proposition 5.1 gives it for K . The counting conclusions therefore follow from the lemma.

Fix $v \notin E^*(t)$. Every preimage is basic and so induces a tuple in T . It cannot induce a tuple in D , since $v \notin F(D)$. The remaining tuple is unique. Thus, writing $Q = p_0 \cdots p_R$, every preimage of v has the form Qz , where $\varphi(z) = d$. It follows that every such preimage is at least $Q\ell(d)$.

For the reverse inequality choose one basic witness with tail w . The deterministic size bound (3.11) gives $\ell(d) \leq w < p_R$. The prefix primes are distinct, so $\gcd(Q, \ell(d)) = 1$ and

$$\varphi(Q\ell(d)) = \varphi(Q)\varphi(\ell(d)) = d \prod_{j=0}^R (p_j - 1) = v.$$

Hence $\ell(v) \leq Q\ell(d)$, proving (5.13). $\square$

The universal quantifier on preimages in Proposition 3.3 is used only after the counting argument, to obtain the common-prefix statement and (5.13). A single basic preimage would give the upper bound for $\ell(v)$ but would not give the reverse inequality.

6 The arithmetic coefficient and the limiting formula

The preceding sections reduce the count of values to distinct tuples with a common long prime prefix. We now sum over the largest prime and replace the remaining prefix by a volume. The shorter tail stays discrete: its different witnesses are combined by a union before taking that volume. Throughout this section, H is fixed first, x tends to infinity next, and H tends to infinity last. In particular, $P = \lfloor \log \log H \rfloor$, $R = m - H$, and $L = m - P$ always refer to the two cuts in (3.1).

6.1 Summing over the largest prime

Let $\mathcal{D}_H(x)$ be the set of distinct data $(d, p_1, \dots, p_R)$ for which there are primes $p_{R+1}, \dots, p_L$ and a positive integer a satisfying the basic conditions (3.10) with p_0 omitted, and

$$d = \varphi(w), \quad w = p_{R+1} \cdots p_L a.$$

The convention $u_0 = B$ is retained in these conditions. Thus the inequality with index zero is a condition on the displayed data and their witness, independent of the eventual choice of p_0 . For any function $f : [1, \infty) \rightarrow [0, 1]$, define

$$M_f(x; H) = \sum_{(d, p_1, \dots, p_R) \in \mathcal{D}_H(x)} \frac{f(\ell(d)/d)}{d \prod_{i=1}^R (p_i - 1)}. \quad (6.1)$$

Each datum is included once, regardless of its number of witnesses. The function $f = 1$ makes the least-preimage ratio irrelevant to this definition.

We use $K_H = \exp(CP\rho^{-P}) = H^{o(1)}$, allowing the absolute constant C to increase in upper bounds. Summing over witnesses, the totient inequality

$$\frac{1}{\varphi(p_{R+1} \cdots p_L a)} \leq \frac{1}{\varphi(a)} \prod_{i=R+1}^L \frac{1}{p_i - 1} \quad (6.2)$$

and Lemma 3.6 give

$$0 \leq M_f(x; H) \leq M_1(x; H) \ll K_H G_m. \quad (6.3)$$

Indeed, the reciprocal sum for the full L prime coordinates is $O(G_m)$ by (3.13), while the unrestricted reciprocal sum for the allowed smooth cofactor is $O(K_H)$ by (3.16). The inequality in (6.2), rather than an equality, also covers the possibility that p_L divides a .

Proposition 6.1 (The unweighted prime sum). *For each fixed $c > 1$, both $t = x$ and $t = x/c$ satisfy*

$$V(t) = \frac{t}{\log x} (M_1(x; H) + E_H(t; x) G_m), \quad \lim_{H \rightarrow \infty} \limsup_{x \rightarrow \infty} \max_{t \in \{x, x/c\}} |E_H(t; x)| = 0. \quad (6.4)$$

All parameters in the two formulas are formed from the same x .

Proof. For fixed data in $\mathcal{D}_H(x)$ put

$$D = d \prod_{i=1}^R (p_i - 1).$$

Since $d \leq w$, the deterministic bound (3.11) implies

$$1 \leq D \leq p_1 \cdots p_L a \leq \exp((\log x)^{0.8}).$$

The possible largest primes are precisely $x^{0.9} \leq p_0 \leq 1 + t/D$. The remaining prime factors are smaller than $x^{0.9}$ for large x , so this choice introduces neither a repeated prefix prime nor another ordering condition. The endpoints t/D lie uniformly in $[x^{1-o(1)}/c, x]$. The prime number theorem, with $\log(t/D) = \log x + o(\log x)$ uniformly, therefore gives

$$\#\{p_0 : x^{0.9} \leq p_0 \leq 1 + t/D\} = \frac{t}{D \log x} (1 + o_{x \rightarrow \infty; H, c}(1)).$$

The lower cutoff costs a relative $O(x^{-0.1+o(1)})$, uniformly in the data. Summing this formula and using (6.3) counts $\mathcal{T}(t)$ with main term $tM_1/\log x$ and error $o_{x \rightarrow \infty; H, c}(xG_m/\log x)$. Proposition 5.3 replaces $\#\mathcal{T}(t)$ by $V(t)$ with an error whose normalized iterated limit is zero. Since t is either x or x/c , this proves (6.4). $\square$

The common mass in Proposition 6.1 already gives the fixed-scaling conclusion. Its proof does not require identifying that mass with the arithmetic coefficient.

Corollary 6.2 (Fixed scaling). *For every fixed real $c > 0$,*

$$\frac{V(cx)}{V(x)} \longrightarrow c.$$

Proof. First let $c > 1$. Apply Proposition 6.1 at $t = x$ and $t = x/c$, using the same $M_1(x; H)$ and $G_m(x)$. Subtracting gives

$$\frac{(V(x) - cV(x/c)) \log x}{xG_m} = E_H(x; x) - E_H(x/c; x).$$

Ford's estimate (3.4) supplies a positive constant c_- with $V(x) \log x / (xG_m) \geq c_-$ for large x . Consequently

$$\limsup_{x \rightarrow \infty} \left| 1 - \frac{cV(x/c)}{V(x)} \right| \leq \frac{1}{c_-} \limsup_{x \rightarrow \infty} (|E_H(x; x)| + |E_H(x/c; x)|).$$

Let H tend to infinity and then replace x by cx to obtain the assertion. This also covers arguments at which $m(x)$ and $m(x/c)$ differ: the latter integer never enters the comparison. Neither continuity of the phase coefficient nor matching values at the ends of the phase interval is required.

The case $c = 1$ is immediate. If $0 < c < 1$, apply the result just proved to the multiplier $1/c > 1$ and the argument $y = cx$; taking reciprocals gives the stated limit. $\square$

6.2 A finite tail and a continuous prefix

The phase-dependent witness set in (2.8) changes with the phase, but for fixed H all its members belong to one finite set. This observation will permit uniform limits without any continuity assumption.

Lemma 6.3 (Uniformly finite tail data). *For each sufficiently large fixed H , the union of all $\mathcal{W}_{H,s}(d)$, over $0 \leq s < 1$ and all d , is finite. All its prime and integer coordinates, and all its totient values, have explicit bounds depending only on H . Uniformly over these data,*

$$0 \leq D(h, \eta) \ll_H \log(2h) \quad (h \geq H). \quad (6.5)$$

Proof. Write $a_* = \lambda/\rho$, so that $\lambda \leq \alpha_s < a_*$. Let

$$q_H = \left\lceil \exp \exp(1.1a_* H \rho^{-H}) \right\rceil, \quad A_H^* = \left\lceil \exp \exp(2a_* P \rho^{-P}) \right\rceil.$$

Every witness belongs to the finite set $\{1, \dots, q_H\}^{H-P} \times \{1, \dots, A_H^*\}$. Its corresponding integer w and totient d satisfy $d \leq w \leq A_H^* q_H^{H-P}$. These bounds do not involve x or s . Finally, $a_j = \int_j^{j+1} \log t \, dt \leq \log(j+1)$, so the definition (2.6) and the bounded tail coordinates give (6.5). $\square$

For $s = \theta(x)$ and $\eta \in \mathcal{W}_{H,s}(d)$, define the region $\mathcal{K}_\eta \subset \mathbb{R}^R$ by $u_0 = B$ and

$$u_i - \sum_{r=i+1}^R a_{r-i} u_r \geq D(m-i, \eta) \quad (0 \leq i \leq R). \quad (6.6)$$

These inequalities imply $u_i \geq 0$. The coordinates u_i here are real; the coarse prime bands have been removed. We write Vol_R for R -dimensional Lebesgue measure.

Lemma 6.4 (The volume of the witness union). *There are numbers $\delta_H \rightarrow 0$ such that, uniformly for $f : [1, \infty) \rightarrow [0, 1]$,*

$$\limsup_{x \rightarrow \infty} \left| \frac{M_f(x; H)}{G_m} - \frac{1}{G_m} \sum_{d: \mathcal{W}_{H,\theta}(d) \neq \emptyset} \frac{f(\ell(d)/d)}{d} \text{Vol}_R \bigcup_{\eta \in \mathcal{W}_{H,\theta}(d)} \mathcal{K}_\eta \right| \leq \delta_H. \quad (6.7)$$

Moreover, for every nonempty finite $T \subseteq \mathcal{W}_{H,\theta}(d)$,

$$\text{Vol}_R \bigcap_{\eta \in T} \mathcal{K}_\eta = G_R \left(1 - \frac{1}{B} \sum_{h=H}^m g_{m-h} \max_{\eta \in T} D(h, \eta) \right)_+^R. \quad (6.8)$$

Here $z_+ = \max(z, 0)$.

Proof. An intersection imposes the largest of the witness thresholds at each slack coordinate.

Intersection volumes. Set $C_i = \max_{\eta \in T} D(m-i, \eta)$ and introduce the prefix slacks

$$t_i = u_i - \sum_{r=i+1}^R a_{r-i} u_r \quad (0 \leq i \leq R).$$

Lemma 3.4 gives $\sum_{i=0}^R g_i t_i = B$, and the substitution from $(u_1, \dots, u_R)$ to $(t_1, \dots, t_R)$ has determinant one. The intersection is $t_i \geq C_i$ for every i , including $i = 0$. Translation by C_i leaves a simplex of available size $B - \sum_i g_i C_i$. Its volume is

$$\frac{(B - \sum_{i=0}^R g_i C_i)_+^R}{R! \prod_{i=1}^R g_i},$$

which proves (6.8) after putting $h = m - i$. It remains to compare the volume of the union with the prime mass.

From primes to volume. Identify $Q_h = p_{m-h}$ for $P \leq h < H$. These are exactly the primes with indices $R+1, \dots, L$. For fixed tail witness η , the original prefix conditions are its coarse bands together with

$$\xi_i u_i - \sum_{r=i+1}^R a_{r-i} u_r \geq D(m-i, \eta) \quad (0 \leq i \leq R). \quad (6.9)$$

Taking the union over $\eta \in \mathcal{W}_{H,\theta}(d)$ is essential: a prime prefix with several witnesses contributes just once to (6.1).

First replace the reciprocal prime sum by the volume of this union with its bands and its ξ_i . In a unit box wholly contained in the union, the product of the prime reciprocal sums differs from its volume by a relative $O(\sum_{i=1}^R e^{-cb_i}) = O(e^{-c'b_R})$, by Lemma 3.6. The geometric growth of b_i as i decreases makes this estimate independent of R . Both the total mass and the total volume before this replacement are $O(K_H G_m)$, by summing over witnesses as in (6.2).

If a unit box meets the union without being contained in it, choose a witness whose region meets the box. That witness's region does not contain the box. Consequently the box lies in a shell at a prefix band endpoint, or within $O(h_i^2)$ of equality in one of (6.9). For an upper bound, sum these shells over witnesses using (6.2). Thicken the discrete tail coordinates into unit boxes as well and sum $1/\varphi(a)$ at a cost $O(K_H)$. The resulting full L -coordinate boxes are covered by the shell bounds of Lemma 3.6 and Proposition 3.8. Their contribution is $o_H(1)G_m$. More explicitly, the slice coefficient for an index with $h = m - i$ is $Lg_i/B \ll \rho^h$. Summing the shell bounds therefore costs at most

$$O\left(K_H \sum_{h \geq H} (he^{-h/40} + h^3 \rho^h)\right) G_m.$$

The broader $O(K_H H^{-2})G_m$ shell estimate in Proposition 3.8 also suffices.

Removing the slack perturbation. Replace ξ_i by 1 for $i \leq R$, retaining the bands. The removed region lies in a shell of thickness $O(e^{-h_i/40} b_i)$, with the same box-thickening errors. Its normalized contribution is at most

$$O\left(K_H \sum_{h \geq H} (he^{-h/40} + h^3 \rho^h)\right) = o_H(1).$$

Both shell estimates are valid for the union because upper bounds may sum over witnesses, while $0 \leq f \leq 1$.

Removing the prefix bands. For every point satisfying the now unperturbed prefix inequalities, recurrence iteration gives

$$u_i \geq g_{R+1-i} v_{H-1} \gg H \rho^{-h_i} \quad (0 \leq i \leq R), \quad (6.10)$$

where $v_{H-1} = \log_2 Q_{H-1}$. To see this, retain the contribution of the first tail coordinate in each prefix inequality, discard the other nonnegative tail contributions, and apply the recurrence for g . The tail band for v_{H-1} and (3.5) give the displayed lower bound. Thickening the $H - P$ tail coordinates changes inequality i by $O(H \log(h_i + 1))$. The enlargement term $Ch_i^2 \rho^{h_i} u_i$ is at least a constant times CHh_i^2 , so it absorbs this error. Together with the retained tail bands, this places the full region inside the enlarged simplex of Lemma 3.6.

To sum these regions without a factor for the number of witnesses, use (6.2) and first fix a . Partition the tail coordinates into half-open unit boxes $\mathcal{C} = \prod_{h=P}^{H-1} \mathcal{C}_h$. Let $\mathcal{U}_{\mathcal{C}}$ be the union of $\mathcal{K}_{\eta}$ over all allowed witnesses $\eta = ((Q_h), a)$ with $(\log_2 Q_h)_h \in \mathcal{C}$. Let E be the set of prefix points violating at least one removed band. Aggregating the reciprocal weights within a box gives

$$\begin{aligned} & \sum_{\substack{\eta=((Q_h), a) \text{ allowed} \\ (\log_2 Q_h)_h \in \mathcal{C}}} \frac{\text{Vol}_R(\mathcal{K}_{\eta} \cap E)}{\prod_h (Q_h - 1)} \\ & \leq \left(\prod_{h=P}^{H-1} \sum_{\log_2 q \in \mathcal{C}_h} \frac{1}{q - 1} \right) \text{Vol}_R(\mathcal{U}_{\mathcal{C}} \cap E) \\ & \ll \text{Vol}_L((\mathcal{U}_{\mathcal{C}} \cap E) \times \mathcal{C}). \end{aligned}$$

The sums over q are over primes. The product is $O(1)$ uniformly in the number of tail coordinates, by (3.12) and their geometrically growing lower endpoints. Here $R + (H - P) = L$ and $\mathcal{C}$ has unit volume. By the preceding enlargement argument, every set on the right lies in the same enlarged L -simplex. Since the tail boxes are disjoint, summing their volumes counts a subset of that simplex once. Finally, summing $1/\varphi(a)$ over the common smooth envelope costs $O(K_H)$. All these bounds hold at each phase; they do not compare witness sets at nearby phases.

Scale that enlarged simplex into the true L -coordinate simplex. Its Jacobian cost is bounded, and its coordinate scaling factors through index R are

$$1 + O\left(\sum_{h \geq H} (e^{-h/40} + h^2 \rho^h)\right) = 1 + o_H(1).$$

For large H , and then large x , a violation of a removed $[0.9\tilde{b}_i, 1.1\tilde{b}_i]$ band therefore becomes a violation of the $[0.95b_i, 1.05b_i]$ band in the true simplex. Lemma 3.5 applies with dimension $L = m - P$, since $h_i \geq H$ and $P/H \rightarrow 0$. Summing its bounds gives $O(e^{-cH})$ of the simplex volume. After the smooth-cofactor factor, the additional volume is at most $O(K_H e^{-cH})G_m = o_H(1)G_m$. This proves (6.7). All constants used in these estimates are uniform in θ , since $\lambda \leq \alpha_{\theta} < \lambda/\rho$ and $\alpha/\alpha_{\theta} = 1 + o_{x \rightarrow \infty; H}(1)$. □

6.3 The arithmetic approximation

Proposition 6.5 (Approximation by the finite coefficient). *There are numbers $\varepsilon_H \rightarrow 0$ such that, for every $f : [1, \infty) \rightarrow [0, 1]$,*

$$\limsup_{x \rightarrow \infty} \left| \frac{M_f(x; H)}{G_m} - A_H(f; \theta(x)) \right| \leq \varepsilon_H. \quad (6.11)$$

The bound is independent of f and of phase. Each $A_H(f; s)$ is a finite sum involving values of f at explicitly bounded rational arguments and absolutely convergent series in its exponential weights, and $A_H(f; s) \geq 0$.

Proof. Apply finite inclusion–exclusion to the union in (6.7), and then use (6.8). For a nonempty subset T of witnesses, write

$$C_T(h) = \max_{\eta \in T} D(h, \eta), \quad S_T = \sum_{h=H}^{\infty} \rho^h C_T(h), \quad q_{m,T} = \frac{1}{B} \sum_{h=H}^m g_{m-h} C_T(h).$$

The series for S_T converges absolutely by (6.5). The finite universe in Lemma 6.3 gives a uniform bound on the number of possible witnesses and their subsets for fixed H ; it also makes all the following estimates uniform in phase, even when membership in a witness set changes.

Using $B = \alpha m \rho^{-m}$ and (3.5), we obtain

$$\begin{aligned} q_{m,T} &= \frac{\gamma}{\alpha m} \sum_{h=H}^m \rho^h C_T(h) + O_H\left(\frac{m \log(2m)}{B}\right) \\ &= \frac{\gamma}{\alpha m} S_T + o_{x \rightarrow \infty; H}(m^{-1}). \end{aligned}$$

The omitted geometric tail is $O_H(\rho^m \log(2m))$. In particular, $q_{m,T} = O_H(1/m)$, and hence

$$(1 - q_{m,T})_+^{m-H} = \exp\left(-\frac{\gamma}{\alpha_\theta} S_T\right) + o_{x \rightarrow \infty; H}(1).$$

Here we used the uniform scale relation $\alpha/\alpha_\theta = 1 + O(\log m/m)$ and $(m-H)q_{m,T}^2 = O_H(1/m)$. The prefactor satisfies

$$\frac{G_{m-H}}{G_m} = \prod_{l=0}^{H-1} \frac{(m-l)g_{m-l}}{B} = \left(\frac{\gamma}{\alpha_\theta}\right)^H \rho^{H(H-1)/2} + o_{x \rightarrow \infty; H}(1). \quad (6.12)$$

There are only boundedly many terms for fixed H . Thus the inclusion–exclusion expression differs from $A_H(f; \theta(x))$ in (2.8) by $o_{x \rightarrow \infty; H}(1)$, uniformly in phase; no witness multiplicity has been introduced. Together with Lemma 6.4, this proves (6.11).

Nonnegativity follows from the finite-union probability interpretation in (2.7): both the union probabilities and the outer weights are nonnegative. $\square$

6.4 Uniform convergence and the asymptotic equivalent

The preceding approximation holds for every bounded nonnegative weight. Convergence as H grows requires a comparison quantity that does not depend on H . For $f = 1$ this quantity is the normalized count of totients. Section 7 will supply a different comparison quantity for each f_k .

Lemma 6.6 (Convergence from a common comparison). *Fix $f : [1, \infty) \rightarrow [0, 1]$. Suppose a real function $C_f(x)$, independent of H , and numbers $\eta_H \rightarrow 0$ satisfy*

$$\limsup_{x \rightarrow \infty} \left| C_f(x) - \frac{M_f(x; H)}{G_m} \right| \leq \eta_H. \quad (6.13)$$

Then $A_H(f; s)$ converges uniformly for $0 \leq s < 1$ to the limit $A(f; s)$ defined in Section 2, and

$$C_f(x) = A(f; \theta(x)) + o(1). \quad (6.14)$$

If additionally $c_- \leq C_f(x) \leq c_+$ for all sufficiently large x , then $c_- \leq A(f; s) \leq c_+$ for every $s \in [0, 1)$.

Proof. By Proposition 6.5,

$$\limsup_{x \rightarrow \infty} |C_f(x) - A_H(f; \theta(x))| \leq \zeta_H, \quad \zeta_H = \eta_H + \varepsilon_H \rightarrow 0.$$

Every phase occurs at arbitrarily large real arguments. Indeed,

$$\psi(B) = \frac{\log B - \log \log B}{\lambda}$$

is continuous and strictly increasing for $B > e$, and tends to infinity. For any $s \in [0, 1)$ solve $\psi(B_n) = n + s$ and put $x_n = \exp(\exp B_n)$. Then $m(x_n) = n$ and $\theta(x_n) = s$ exactly. Comparing indices H and K along this same sequence gives

$$|A_H(f; s) - A_K(f; s)| \leq \zeta_H + \zeta_K.$$

This bound is independent of s , so the functions form a uniformly Cauchy sequence. In particular, $\sup_s |A_H(f; s) - A(f; s)| \leq \zeta_H$. Combining this inequality with the preceding limsup bound and then sending H to infinity proves (6.14). If C_f lies between c_- and c_+ , apply those bounds on each exact-phase sequence and let H grow to obtain the asserted bounds for A . No continuity in the phase was used. $\square$

Proof of Theorem 2.1. Set

$$C_1(x) = \frac{V(x) \log x}{x G_m}.$$

Proposition 6.1 with $t = x$ supplies (6.13). Lemma 6.6 therefore proves uniform existence of $A(1; s)$ and $C_1(x) = A(1; \theta(x)) + o(1)$. Ford's two-sided estimate (3.4) supplies positive absolute constants c_-, c_+ bounding C_1 . The same lemma gives

$$0 < c_- \leq A(1; s) \leq c_+ < \infty \quad (0 \leq s < 1).$$

The additive approximation is consequently equivalent to

$$V(x) \sim \frac{x}{\log x} G_m A(1; \theta(x)).$$

The coefficient in this formula was defined by the finite arithmetic expressions (2.8); the count V has been used to establish their convergence and bounds, not to define them.

Corollary 6.2 supplies the remaining assertion of the theorem for every $c > 0$. $\square$

7 Least preimages

We now prove Theorem 2.2. The prime that dominates a basic representation also determines the interval in which its least preimage lies. Counting that prime first gives the weight f_k . The resulting approximation is additive; a separate argument will show when its coefficient is bounded away from zero.

Throughout this section, $k \geq 1$ is a fixed integer. The mass $M_f(x; H)$ is defined in (6.1); its summands correspond to distinct data $(d, p_1, \dots, p_R)$, regardless of how many basic witnesses yield the same data. Write

$$f_k(r) = \begin{cases} 0, & 1 \leq r \leq k, \\ 1 - k/r, & k < r < k + 1, \\ 1/r, & r \geq k + 1. \end{cases}$$

Thus $0 \leq f_k \leq 1/(k + 1)$, and $f_k(r) > 0$ exactly when $r > k$. Equivalently, $f_k(r)$ is the length of

$$\{t \in [0, 1] : k < rt \leq k + 1\}.$$

It is therefore the permitted fraction of the normalized value interval when the least-preimage ratio is fixed at r .

7.1 Counting the interval for the largest prime

Lemma 7.1. *For each sufficiently large fixed H , as x tends to infinity,*

$$\left| N_k(x) - \frac{x}{\log x} M_{f_k}(x; H) \right| \leq (\epsilon_{H,k} + o_{x \rightarrow \infty; H, k}(1)) \frac{x G_m}{\log x}, \quad (7.1)$$

where $\epsilon_{H,k} \geq 0$ tends to zero as $H \rightarrow \infty$.

Proof. Proposition 5.3 allows us, with an error of $(\epsilon_H + o_{x \rightarrow \infty; H}(1)) x G_m / \log x$, to count tuples $(p_0, \dots, p_R, d)$ satisfying

$$(p_0 - 1) \cdots (p_R - 1) d \leq x, \quad kx < p_0 \cdots p_R \ell(d) \leq (k+1)x.$$

Both the exceptional values and the tuples above them are controlled there. This use of the least-preimage identity requires that every preimage have the same prefix. The identity would not follow from the existence of a single basic witness.

Fix the remaining data $(d, p_1, \dots, p_R)$, and put

$$D = d \prod_{i=1}^R (p_i - 1), \quad A = \ell(d) \prod_{i=1}^R p_i, \quad r = \frac{\ell(d)}{d}, \quad q = \prod_{i=1}^R \left(1 - \frac{1}{p_i}\right).$$

Here $r \geq 1$. Set $T = x/D$ and $U = x/A = Tq/r$. The admissible largest primes satisfy exactly

$$p_0 \geq x^{0.9}, \quad kU < p_0 \leq \min\{T+1, (k+1)U\}. \quad (7.2)$$

Once the lower bound for p_0 holds, the basic-witness restrictions on the remaining data do not change with p_0 ; recall that their zeroth simplex coordinate is B .

Ignoring the lower cutoff and replacing $T+1$ by T , the interval has length

$$\min\{T, (k+1)U\} - \min\{T, kU\} = T f_k(r/q).$$

The replacement changes its length by at most one. Moreover, $z \mapsto f_k(1/z)$ is $(k+1)$ -Lipschitz on $[0, 1]$, where its value at zero is zero. Consequently

$$|f_k(r/q) - f_k(r)| \leq \frac{(k+1)(1-q)}{r} \leq (k+1) \sum_{i=1}^R \frac{1}{p_i}. \quad (7.3)$$

For fixed H , the deterministic size bounds in (3.11) give $T = x^{1-o(1)}$ uniformly in the remaining data. The tail has a witness w bounded in terms of H only, so $\ell(d) \leq w$ bounds r in terms of H . The prime bands bound q away from zero. Hence $U = x^{1-o(1)}$ uniformly as well, and $kU > x^{0.9}$ for all sufficiently large x . The lower cutoff in (7.2) is then inactive.

Apply the prime number theorem at its two endpoints, taking the positive part of the difference if the interval is empty. Each nonzero endpoint is $x^{1-o(1)}$, and each is $O_k(T)$. Thus this gives, uniformly in the remaining data,

$$\#\{p_0 \text{ satisfying (7.2)}\} = \frac{T}{\log x} \left(f_k(r) + O_k \left(\sum_{i=1}^R \frac{1}{p_i} \right) + o_{x \rightarrow \infty; H, k}(1) \right).$$

The shift by one is included in the last error. This is an additive estimate relative to $T/\log x$, obtained from the ordinary prime number theorem; no relative estimate on a short prime interval is required.

The geometric separation of the bands implies

$$\sup \sum_{i=1}^R \frac{1}{p_i} \ll \exp(-cH\rho^{-H}).$$

Indeed $b_{i-1}/b_i \geq \rho^{-1}$, and $p_i \geq \exp \exp(cb_i)$; the resulting reciprocal series is bounded by a constant times its smallest-scale bound. In particular, this estimate is uniform in the growing number R of primes. Since $M_1 \ll K_H G_m$ and $K_H = H^{o(1)}$, summing the displayed prime count gives an error bounded by

$$\left(O_k(K_H \exp(-cH\rho^{-H})) + o_{x \rightarrow \infty; H, k}(1) \right) \frac{x G_m}{\log x}.$$

Together with the tuple-to-value error this proves (7.1). $\square$

The additive form is necessary at this stage. For example, when $r = k$ and $q < 1$, one has $f_k(r) = 0$ but $f_k(r/q) > 0$. Inequality (7.3) controls this boundary without asserting a relative approximation to a zero term.

Apply Lemma 6.6 with the function

$$C_{f_k}(x) = \frac{N_k(x) \log x}{x G_m},$$

which is independent of H . Lemma 7.1 provides its required comparison with M_{f_k}/G_m . We obtain the uniform convergence of $A_H(f_k; s)$ on $0 \leq s < 1$ and

$$N_k(x) = \frac{x G_m}{\log x} (A(f_k; \theta) + o(1)). \quad (7.4)$$

It remains to distinguish a coefficient bounded away from zero from an identically zero count.

7.2 Seed propagation and bounded ratios

The positivity argument needs two distinct facts. A seed with $\ell(d)/d > k$ must produce a positive proportion of values whose ratios stay away from k . These ratios must also remain bounded on a positive proportion of values, since $f_k(r) \rightarrow 0$ as $r \rightarrow \infty$. The next two lemmas supply these facts.

Preservation of a complete inverse fiber under multiplication by a prime already appears in Erdős's proof of Theorem 4 [3]. Ford's construction gives such preservation on a positive proportion of the distinct-totient scale. A later simultaneous two-seed form appears in Pollack, Pomerance and Treviño [13, Lemma 4.1], again using Ford's construction. We use the following single-seed consequence of Ford's proof.

Lemma 7.2 (Ford's inverse-fiber propagation). *Fix a totient d , and write $\varphi^{-1}(d) = \{d_1, \dots, d_\kappa\}$. There are constants $\eta_d > 0$ and x_d such that, for every $x \geq x_d$, at least $\eta_d V(x)$ distinct totients $v \leq x$ have the form*

$$v = d\varphi(b), \quad \varphi^{-1}(v) = \{bd_1, \dots, bd_\kappa\}$$

for some positive integer b . In particular, these values satisfy $\ell(v)/v \geq \ell(d)/d$.

Proof. Use the construction in the proof of Theorem 2 of [7, Section 5, pp. 24–26, Equations (5.10)–(5.19)]. For the fixed seed d , Ford constructs a set $\mathcal{B}$ and excludes at most half its members. For each remaining b , every preimage of $d\varphi(b)$ is one of the products bd_i . The paragraph following Equation (5.18) also proves that these members give distinct totients. The concluding bound on p. 26 is $|\mathcal{B}|/2 \gg_\varepsilon d^{-1-\varepsilon}V(x)$ for sufficiently large x . This full-fiber conclusion is recorded explicitly in [7, Section 7.3, p. 39]. Finally,

$$\frac{\ell(v)}{v} = \frac{b}{\varphi(b)} \frac{\ell(d)}{d} \geq \frac{\ell(d)}{d}.$$

□

We also need to keep these ratios in a bounded interval. This follows from the existence of a bounded arithmetic tail and does not require uniqueness of the prefix.

Lemma 7.3. *For every $\varepsilon > 0$, there are $C < \infty$ and x_0 such that*

$$\#\{v \in \mathcal{V} : v \leq x, \ell(v)/v > C\} \leq \varepsilon V(x) \quad (x \geq x_0).$$

Proof. Choose one sufficiently large fixed cut H_0 in Proposition 3.3, so that its exceptional set has at most $\varepsilon V(x)$ members for all sufficiently large x . Every remaining value has a basic witness $n = p_0 \cdots p_{R_0} w$, where $R_0 = m - H_0$, $w < p_{R_0}$, and w is bounded in terms of H_0 only. The prefix primes are distinct, and their bands give a uniform bound for $\prod_{i=0}^{R_0} p_i/(p_i - 1)$. Therefore

$$\frac{\ell(v)}{v} \leq \frac{n}{\varphi(n)} = \frac{w}{\varphi(w)} \prod_{i=0}^{R_0} \frac{p_i}{p_i - 1} \leq C(H_0).$$

The last bound is independent of x , as required. □

7.3 The positive and zero alternatives

Suppose first that there is a totient d_* with $\ell(d_*) > kd_*$. Choose $\delta > 0$ such that $\ell(d_*)/d_* > k + 2\delta$. Lemma 7.2 gives a fixed $\eta > 0$ such that at least $\eta V(x)$ values satisfy $\ell(v)/v > k + 2\delta$ for every sufficiently large x . Apply Lemma 7.3 with $\varepsilon = \eta/4$. It supplies a fixed C for which at least $3\eta V(x)/4$ values satisfy

$$k + 2\delta < \frac{\ell(v)}{v} \leq C. \tag{7.5}$$

The numbers η, δ, C are fixed before the cut H is allowed to grow.

For every sufficiently large H , and then sufficiently large x , Proposition 5.3 removes at most $\eta V(x)/4$ of these values. For each remaining value its unique tuple satisfies

$$\frac{\ell(v)}{v} = \frac{\ell(d)}{d} \prod_{i=0}^R \frac{p_i}{p_i - 1}.$$

The product is $1 + O(\epsilon_H) + o_{x \rightarrow \infty; H}(1)$, uniformly in the tuple. Increasing H and then x , we may bound it by $(k + 2\delta)/(k + \delta)$. Thus at least $\eta V(x)/2$ distinct tuples have their tail ratio in the fixed interval

$$I = [k + \delta, C].$$

Let $M_{\mathbf{1}_I}$ denote the mass (6.1) with weight $\mathbf{1}_I(r)$. The ordinary prime number theorem, counting all admissible largest primes for data whose tail ratio lies in I , bounds the number of such tuples above by

$$(1 + o_{x \rightarrow \infty; H}(1)) \frac{x}{\log x} M_{\mathbf{1}_I}(x; H).$$

Comparison with $\eta V(x)/2$, followed by (3.4), gives

$$M_{\mathbf{1}_I}(x; H) \geq c_1 G_m$$

for all sufficiently large H and then sufficiently large x , where $c_1 > 0$ is independent of H . Since

$$\inf_{r \in I} f_k(r) \geq \min \left\{ \frac{\delta}{k + \delta}, \frac{1}{C} \right\} > 0,$$

we conclude that

$$M_{f_k}(x; H) \geq c_2 G_m \tag{7.6}$$

with $c_2 > 0$ independent of sufficiently large H .

We transfer this lower bound first to the normalized count $C_{f_k}(x) = N_k(x) \log x / (x G_m)$, which does not depend on H . Fix a sufficiently large H such that the error $\epsilon_{H,k}$ in (7.1) is less than $c_2/4$, and then take x large enough that its remaining error is less than $c_2/4$. Equations (7.1) and (7.6) give $C_{f_k}(x) \geq c_2/2$ for every sufficiently large x . This count is also bounded above, since $N_k(x) \leq V(x)$ and (3.4) holds. The bounds clause of Lemma 6.6 therefore gives

$$\inf_{0 \leq s < 1} A(f_k; s) \geq c_2/2 > 0.$$

Therefore the additive formula (7.4) is multiplicative in this case. Its positive lower bound, the inequality $N_k(x) \leq V(x)$, and (3.4) give $N_k(x) \asymp_k V(x)$.

If instead $\ell(d) \leq kd$ for every totient d , then $\ell(v) \leq kv \leq kx$ for every $v \leq x$. Hence $N_k(x) = 0$ for every x . Every weight $f_k(\ell(d)/d)$ in the finite arithmetic formula is also zero, so $A_H(f_k; s) = A(f_k; s) = 0$. This proves the dichotomy in Theorem 2.2. Its final assertion is established next.

Remark 7.4. The positive alternative occurs for $k = 1$ and $k = 2$. Ford gives $d = 2^{18} \cdot 257$ as a totient all of whose preimages are divisible by 8 [7, Section 7.3, p. 39]. Each such preimage n is even and has an odd prime factor: a power of 2 would have a power of 2 as its totient. Thus $n/\varphi(n) = \prod_{p|n} p/(p-1) > 2$, and $\ell(d) > 2d$.

We do not determine all integers k for which the positive alternative occurs. Its occurrence for every integer $k \geq 1$ is equivalent to the unboundedness of $\ell(d)/d$ over totients d . Neither that unboundedness nor the complete classification of k is established here.

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